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Synthesis and interaction with growth factors of sulfated oligosaccharides containing an anomeric fluorinated tail

José L. de Paz [a,*], María José García-Jiménez [a], Vahid Jafari [b], Mario García-Domínguez [b], Pedro M. Nieto [a,*]

[a] Glycosystems Laboratory, Instituto de Investigaciones Químicas (IIQ), cicCartuja, CSIC and Universidad de Sevilla, Americo Vespucio, 49, 41092 Sevilla, Spain
[b] Andalusian Center for Molecular Biology and Regenerative Medicine-CABIMER, CSIC-Universidad de Sevilla-Universidad Pablo de Olavide, Américo Vespucio, 24, 41092 Sevilla, Spain

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ABSTRACT

Compounds that mimic the biological properties of glycosaminoglycans (GAGs) and can be more easily prepared than the native GAG oligosaccharides are highly demanded. Here, we present the synthesis of sulfated oligosaccharides displaying a perfluorinated aliphatic tag at the reducing end as GAG mimetics. The preparation of these molecules was greatly facilitated by the presence of the fluorinated tail since the reaction intermediates were isolated by simple fluorous solid-phase extraction. Fluorescence polarization competition assays indicated that the synthesized oligosaccharides interacted with two heparin-binding growth factors, midkine (MK) and FGF-2, showing higher binding affinities than the natural oligosaccharides, and can be therefore considered as useful GAG mimetics. Moreover, NMR experiments showed that the 3D structure of these compounds is similar to that of the native sequences, in terms of sugar ring and glycosidic linkage conformations. Finally, we also demonstrated that these derivatives are able to block the MK-stimulating effect on NIH3T3 cells growth.

1. Introduction

Glycosaminoglycans (GAGs) are a family of sulfated, highly heterogeneous polysaccharides that regulate multiple biological processes [1]. Heparin, heparan sulfate, chondroitin sulfate and hyaluronic acid are members of this family. It is known that GAG biological activities are usually triggered by the interaction between particular, structurally homogeneous oligosaccharides sequences and certain proteins [2]. The chemical synthesis of these GAG oligomers is, however, a very challenging and arduous task [3]. The development and use of GAG mimetics [4–10], more easily prepared than GAG oligosaccharides, is an interesting alternative to modulate GAG-protein interactions. In this context, sulfated oligosaccharides, presenting simpler and more accessible structure compared to GAG natural oligomers, have been reported as GAG mimetics [11–19]. For instance, Kikkeri and coworkers recently reported the synthesis of IdoA homo-oligosaccharides with different sulfate distributions and demonstrated their ability to interact with chemokines and attenuate cancer progression [20]. Pixatimod (PG545) is another sulfated oligosaccharide that imitates the binding properties of GAG sequences [21]. Thus, this clinical-stage mimetic is able to interact with SARS-Cov2 Spike protein, blocking its binding to the ACE2 receptor.

We have previously synthesized several sulfated oligosaccharides that strongly bound to heparin-binding growth factors such as midkine (MK) and basic fibroblast growth factor (FGF-2) [22,23]. In these compounds, all the hydroxyls were masked with hydrophobic protecting groups since we previously found that the protein binding affinities of sulfated, protected oligosaccharides were much higher than those corresponding to the fully deprotected ones [24]. However, these derivatives displayed multiple labile ester linkages and this fact limited their potential biological applications.

Here, we present the synthesis of oligosaccharides **1** and **2**, following the sequence GlcN (4,6-di-OSO_3)-β(1 → 4)-Glc-β(1 → 3), closely related to the structure of biologically active chondroitin sulfate E [25–27] (CS-E, Fig. 1). Naturally occurring GalN and GlcA units were replaced by GlcN and Glc moieties, respectively, in order to facilitate the synthetic access to these GAG-like oligosaccharides. Thus, for the synthesis of CS-E oligosaccharides, GalN building blocks containing a cyclic protecting group at positions 4 and 6 are usually employed [28–33]. However, this type of protection can lead to anomeric mixtures in the glycosylations

* Corresponding authors.
E-mail addresses: jlpaz@iiq.csic.es (J.L. de Paz), pedro.nieto@iiq.csic.es (P.M. Nieto).

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[34–38]. The use of GlcN units avoids this problem. On the other hand, GlcA moieties are considered as low reactive building blocks in glycosylation reactions. In addition, the combined use of GlcA and *N*-phthaloyl GlcN building blocks is sometimes associated with problems in the deprotection steps [39]. The use of Glc rather than GlcA solves these limitations. Importantly, the presence of Glc instead of GlcA in CS-E like derivatives did not significantly change the 3D structure and the binding properties of the oligosaccharides [40]. Moreover, compounds **1** and **2** contain benzyl groups at position 3 of the Glc units because we previously reported that the presence of these groups increased the binding affinities of the oligosaccharides without a significant change in their 3D structure [41].

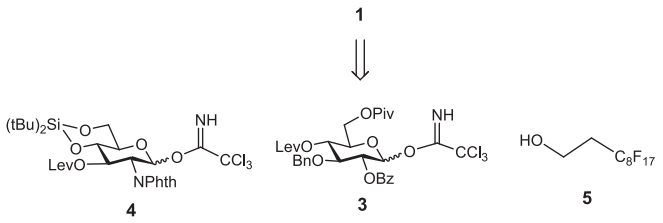
Oligosaccharides **1** and **2** also contain a fluorinated tag at the reducing end anomeric position. The introduction of the C_8F_{17} tail served several purposes. First, the synthesis of the oligosaccharides was greatly facilitated because the reaction intermediates could be purified by simple fluorous solid-phase extraction (F-SPE) [42,43]. Moreover, we hypothesized that the presence of the fluorous tag could enhance the strength of the binding of our oligosaccharides for a protein. In fact, previous studies have shown that the attachment of a hydrophobic group (such as a fluorocarbon tail) to a ligand increased protein binding [44]. This fact could be explained by the energetically favoured desolvation of the hydrophobic fluorinated surface in combination with additional dipolar interactions of the C-F bonds with the surface of the protein receptor (the so-called polar hydrophobicity effect) [45]. In addition, multiple papers have shown the interesting properties of (poly) fluorinated carbohydrate derivatives [46–49], and the advantages of incorporating fluorine atoms in bioorganic and medicinal chemistry [50,51], such as improved binding interactions, bioavailability, metabolic stability and cell membrane permeability.

With the target oligosaccharides in hand, we measured their binding to two proteins, MK and FGF-2, by using fluorescence polarization competition assays. Saturation transfer difference (STD) NMR experiments confirmed the interaction between derivative **1** and MK. Finally, we also tested the effect of the synthetic compounds on the MK-mediated proliferation of NIH3T3 cells.

2. Results and discussion

2.1. Chemical synthesis

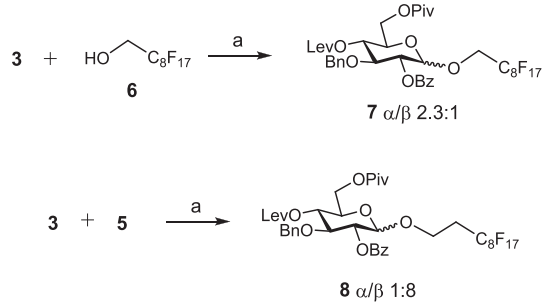
For the preparation of tetrasaccharide **1**, we employed trichloroacetimidate building blocks **3** [23] and **4** [22], and commercially available perfluorinated alcohol acceptor **5** as starting materials (Scheme 1). Levulinoyl (Lev) groups were chosen for the temporary protection of the hydroxyl groups that would be glycosylated. On the other hand, benzoyl (Bz) and *N*-phthaloyl (*N*-Phth) participating groups at position 2 of the monosaccharides allowed the selective formation of the desired 1,2-*trans* glycosidic bonds. Moreover, the cyclic silylene



Scheme 1. Starting materials for the synthesis of tetrasaccharide **1**. Piv = pivaloyl.

group was installed at positions 4 and 6 of the GlcN moiety for selective deprotection and further sulfation at the late stages of the synthesis.

In order to introduce the fluorinated tail at the anomeric position of Glc reducing end, we first attempted the condensation between donor **3** and commercially available perfluorinated nonanol **6** (Scheme 2). The reaction was performed at room temperature, using *tert*-butyldimethylsilyl trifluoromethanesulfonate (TBSOTf) as catalyst and CH_3CN as solvent, due to the insolubility of acceptor **6** in CH_2Cl_2. After F-SPE purification, two spots were detected by TLC in the fluorophilic fraction. NMR data unambiguously indicated the formation of a mixture of α/β anomers in a 2.3:1 ratio, in spite of the presence of a *C*2-participating group. It is well known that the nucleophilicity of the glycosyl acceptor influences on the stereochemistry of a glycosylation reaction [52,53]. Recent studies have demonstrated this fact by using model alcohol acceptors with different nucleophilicities due to the presence of variable amounts of fluorine atoms [54,55]. It has been proposed that the glycosylation of weak nucleophiles follow a reaction mechanism with more S_N1 character, giving preferentially the α glycoside. On the contrary, more reactive glycosyl acceptors favour the formation of the β anomer, through a S_N2-like mechanism [55]. This can explain the stereochemical outcome of the condensation between **3** and **6**: the



Scheme 2. Reagents and conditions: a) TBSOTf, CH_3CN, room temperature, 55 % (**7**), 85 % (**8**). Yields were calculated after purification of the α/β mixture by F-SPE. The ratio of α/β anomers was determined by integration of diagnostic peaks in the ^1H NMR spectra.

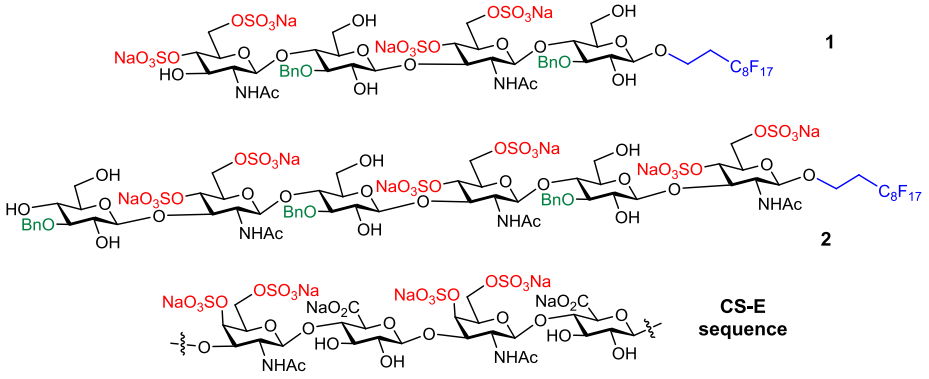


Fig. 1. Structure of tetrasaccharide **1** and hexasaccharide **2**. For comparison purposes, a natural CS-E sequence (sodium salt) is also shown.

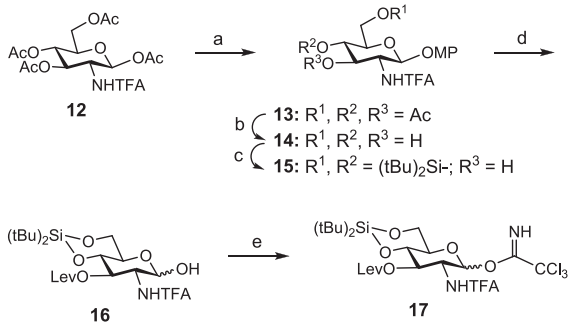
extremely low nucleophilicity of the fluorinated alcohol led to the preferred formation of the α anomer, even in the presence of a *C*2-participating group, by preferential reaction with more reactive oxocarbenium ion intermediate in a S_N1-like substitution [56].

To improve the stereoselectivity and yield, we then probed the glycosylation of **3** and perfluorinated decanol **5**, bearing an additional CH_2 between the fluorine atoms and the hydroxyl function that should increase the nucleophilicity of the acceptor. Remarkably, the stereoselectivity was reversed and the β glycoside was obtained as the major product (Scheme 2). These results show that just the addition of a methylene group in the acceptor have a great impact on the glycosylation stereoselectivity.

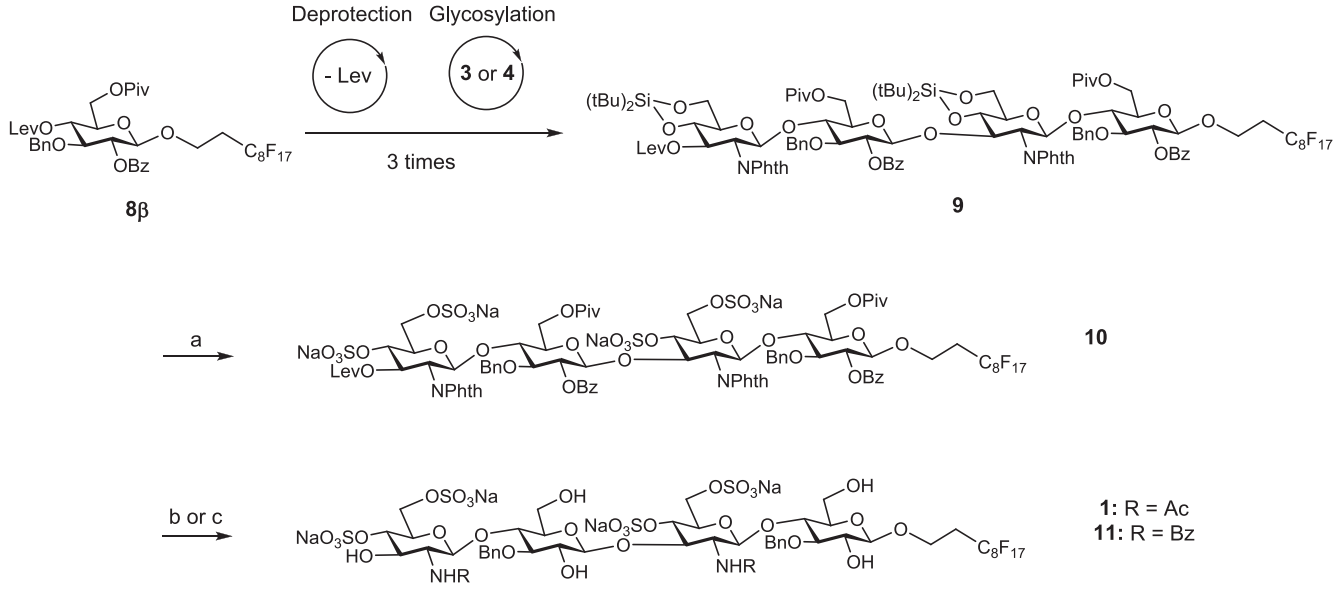
Once the perfluorinated tag was introduced at the anomeric position in high yield and with good stereoselectivity, we carried out the assembly of the tetrasaccharide chain (Scheme 3). The β anomer of monosaccharide **8** was first purified by standard silica gel chromatography. Then, this compound was submitted to a sequence of Lev deprotection reactions, releasing the acceptor hydroxyl group, and glycosylations with trichloroacetimidate building blocks **4** and **3**. The glycosylation with donor **3** required the use of a higher amount of TBSOTf (20 mol% with respect to the donor) to avoid the formation of orthoester side products. All the reaction intermediates were isolated by simple F-SPE. TLC analysis of the fluorine containing fraction revealed a single main spot after each transformation, and mass spectrometry data confirmed the formation of the desired intermediates (see Materials and Methods section). Noteworthy, only 2–3 equivalents of trichloroacetimidate donors were required to complete the couplings and consume the glycosyl acceptor, underlining the advantages of the solution-phase approaches for oligosaccharide synthesis in comparison to solid-phase strategies. Thus, pure tetrasaccharide **9** was obtained in excellent 48 % yield from **8**, after 6 steps (89 % average yield per step).

The cyclic silylene groups were then selectively deprotected and the obtained tetraol was sulfated under microwave irradiation to give **10** as sodium salt after purification by Sephadex LH-20 and silica gel chromatography, and treatment with Dowex Na^+ ion-exchange resin. Deprotection of *N*-Phth groups was accomplished by using ethylenediamine at 120 °C under microwave heating. Further treatment with an aqueous solution of NaOH (4 M) was required to fully deprotect the acyl protecting groups. Finally, selective *N*-acetylation was performed using acetic anhydride and Et_3N in MeOH to afford the target tetrasaccharide **1** in 78 % yield. Alternatively, the amine intermediate was selectively *N*-benzoylated by using benzoic anhydride and Et_3N in MeOH to give **11**. This derivative would allow us to evaluate the impact of the two additional phenyl rings on the protein binding affinities.

For the synthesis of hexasaccharide **2**, we decided to explore the use of a different GlcN unit, displaying a *N*-trifluroacetyl (*N*-TFA) group (Scheme 4). The *N*-TFA group also favours the formation of the β glycosidic bonds but can be deprotected under milder conditions, compared with the *N*-Phth moiety. The preparation of this new building block started from peracetylated *N*-trifluoroacetyl-protected β-galactosamine derivative **12** [57]. This compound was efficiently transformed into the corresponding 4-methoxyphenyl β glycoside **13** [58] by reaction with 4-methoxyphenol catalized by trimethylsilyl trifluoromethanesulfonate (TMSOTf) at 0 °C. Treatment with NaOMe in MeOH (→**14** [58]) followed by reaction with di-*tert*-butylsilyl bis(trifluoromethanesulfonate) (tBu$_2$Si(OTf)$_2$) in dry pyridine afforded compound **15** containing the orthogonal silyl protecting group for further selective sulfation. Lev group was then installed at position 3 using levulinic anhydride and DMAP as catalyst. Oxidative removal of the 4-methoxyphenyl group with cerium (IV) ammonium nitrate (CAN) at 0 °C afforded hemiacetal **16** in good yield. Finally, derivative **16** was



Scheme 4. Reagents and conditions: a) 4-methoxyphenol, TMSOTf, CH_2Cl_2, 0 °C, 79 %; b) NaOMe, MeOH, 98 %; c) tBu$_2$Si(OTf)$_2$, Py, 0 °C → room temperature, 70 %; d) Lev$_2$O, DMAP, CH_2Cl_2, 97 %; CAN, CH_2Cl_2/CH_3CN/H_2O, 0 °C, 84 %; e) Cl_3CCN, DBU, CH_2Cl_2, 94 %. MP = 4-methoxyphenyl; DMAP = 4-dimethylamino pyridine; DBU = 1,8-diazabicycloundec-7-ene.



Scheme 3. Reagents and conditions: Lev deprotection: $NH_2NH_2{\cdot}H_2O$ (2 equiv.), Py/AcOH, CH_2Cl_2; Glycosylation: TBSOTf, CH_2Cl_2, room temperature; 48 % (**9**); a) $(HF)_n$·Py, THF, 0 °C; $SO_3{\cdot}Me_3N$, DMF, 100 °C, MW heating, 72 %; b) $NH_2(CH_2)_2NH_2$, *n*-BuOH/CF_3CH_2OH, 120 °C, MW heating; NaOH, MeOH/CF_3CH_2OH; Ac_2O, Et_3N, MeOH, 78 % (**1**); c) $NH_2(CH_2)_2NH_2$, *n*-BuOH/CF_3CH_2OH, 120 °C, MW heating; NaOH, MeOH/CF_3CH_2OH; Bz_2O, Et_3N, MeOH, 78 % (**11**).

activated as a trichloroacetimidate donor (→**17**) by treatment with Cl_3CCN and catalytic DBU in dry CH_2Cl_2. Thus, compound **17** was available for testing in glycosylation reactions.

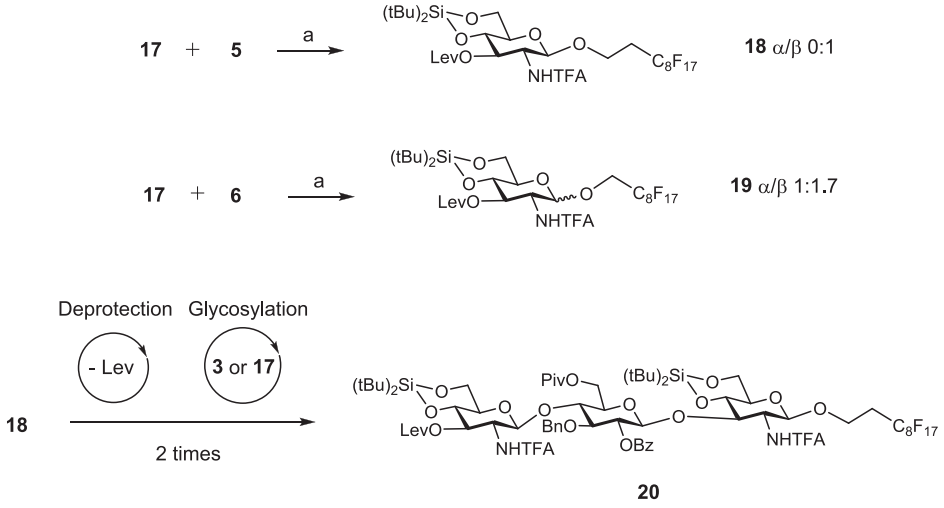
For the introduction of the fluorous tag at the anomeric position of the GlcN unit, we run the glycosylation between donor **17** and alcohol **5** (Scheme 5). In this case, the β glycoside **18** was exclusively obtained and no α anomer was detected. For comparison purposes, we also decided to perform the coupling with acceptor **6**. In spite of the presence of the *N*-TFA participating group, a mixture of α/β anomers, in a 1:1.7 ratio, was isolated. This result confirmed that the unexpected α product can be formed when decreasing the acceptor nucleophilicity.

Next, we studied the performance of the *N*-TFA building blocks in the fluorous-assisted oligosaccharide assembly. For this purpose, monosaccharide **18** was submitted to the Lev deprotection and glycosylation conditions (Scheme 5). Disappointingly, when we tried the synthesis of trisaccharide **20** by glycosylation between trichloroacetimidate **17** and the corresponding 4-OH Glc-GlcN disaccharide, an important amount (>50 %) of unreacted glycosyl acceptor was detected. A second reaction cycle with additional donor **17** was required to decrease the remaining acceptor and obtain compound **20** in 46 % yield, after 4 steps (83 % average yield per step). This result pointed out that a high excess of glycosyl donor **17** (5 equiv.) was needed to complete the glycosidation reaction and obtain trisaccharide **20** in good yield. Previous reports also indicated a reduced performance of *N*-TFA building blocks in certain glycosylation reactions due to the formation of oxazoline byproducts [34] and the participation of the amide NH in hydrogen bonds involving the oxygen atom of the acceptor hydroxyl group [59,60]. Therefore, the *N*-TFA trichloroacetimidate **17** seemed to be less convenient than the *N*-Phth analogue for the preparation of long oligomers, such as hexasaccharide **2**. For this reason, we turned our attention to the *N*-Phth protected donor **4**.

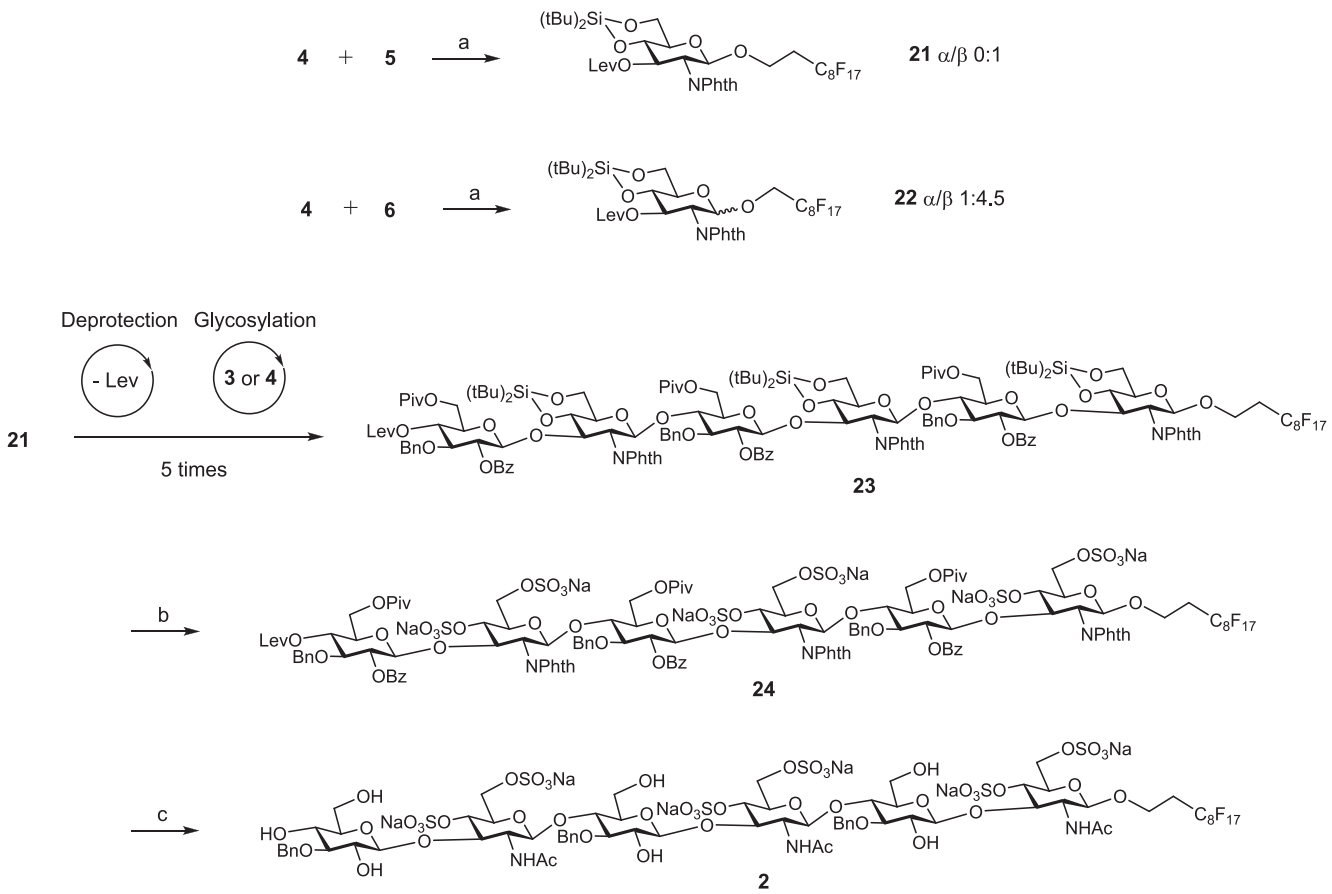
Thus, the fluorinated tail was introduced at the anomeric position of the reducing end GlcN unit by glycosylation between trichloroacetimidate **4** and alcohol **5** (Scheme 6). The desired β glycoside was exclusively obtained in 78 % yield without detecting any α anomer. Again, the glycosylation with low reactive fluorinated nonanol increased the α-glycoside formation, although to a minor extent (1:4.5 α/β ratio). Monosaccharide **21** was then submitted to the Lev deprotection and glycosylation reaction sequence. After F-SPE purification, the TLC and mass spectrometry analysis of the fluorous containing fraction of each intermediate showed the formation of the desired compound as major product. Importantly, just 2–3 equivalents of glycosyl donor were required to complete the coupling steps. In this way, hexasaccharide **23** was obtained uneventfully in excellent 24 % yield after 10 steps (87 % average yield per step). The selective deprotection of the cyclic silylene groups afforded the corresponding hexaol that was treated with $SO_3{\cdot}NMe_3$ complex in DMF at 100 °C under microwave irradiation (Scheme 6). This sulfation reaction proved to be quite challenging. TLC analysis revealed the formation of a mixture of partially sulfated compounds, which was purified by F-SPE using, in this case, 40:60 MeOH/H_2O for the fluorophobic elution and 100 % MeOH for the fluorophilic one [61]. Two additional reaction cycles at 120 °C were performed to increase the ratio of the fully sulfated hexamer **24**, which was finally isolated as sodium salt after silica gel column chromatography and Dowex Na^+ treatment. The *N*-Phth groups were removed by using ethylene diamine in a *n*-BuOH/CF_3CH_2OH mixture at 120 °C (microwave heating) and the Piv and Bz moieties were fully deprotected by further treatment with NaOH. The released amino groups were then selectively acetylated and the target hexasaccharide **2** was finally isolated in good yield after silica gel chromatography and ion-exchange resin treatment.

2.2. Fluorescence polarization competition experiments

After accomplishing the synthesis of the target oligosaccharides, we studied their interactions with MK and FGF-2, two proteins with a relevant role in cell proliferation and cáncer [62,63]. We chose these two proteins as models of GAG-binding growth factors: while MK binds with high affinity to both heparin and CS-E, FGF-2 shows a preferential binding to heparin compared with CS-E [25,64]. Since the synthesized oligosaccharides mimic CS-E, we expected higher binding affinities for MK. We employed a fluorescence polarization competition experiment previously developed in our lab (for details, see Materials and Methods section) [33,65]. This assay provided relative binding affinities for each synthetic oligosaccharide, expressed as IC_{50} values (Table 1). Tetrasaccharide **1** and hexasaccharide **2** bound to MK and FGF-2 in the micromolar range. As expected, longer derivative **2** showed higher binding affinities compared to **1**, for both proteins. Interestingly, the binding affinities of tetrasaccharide **11** displaying benzamido instead of acetamido groups were higher than those showed by **1**, particularly when considering MK (IC_{50} = 4.2 μM against 32 μM). These data suggest that increasing the overall hydrophobicity by the presence of additional phenyl rings enhances the binding strength, in good agreement with our previous results [24]. In fact, sulfated intermediate **10** gave the lowest IC_{50} values, in the nanomolar range for MK. However, this type of compounds has a reduced stability for biological use due to the presence of numerous ester linkages. In addition, all the tested compounds presented a slight degree of selectivity for MK over FGF-2, with lower IC_{50}



Scheme 5. Reagents and conditions: a) TBSOTf, CH_3CN, room temperature, 68 % (**18**), 46 % (**19**). The ratio of α/β anomers was determined by integration of diagnostic peaks in the ^1H NMR spectra. Lev deprotection: $NH_2NH_2{\cdot}H_2O$ (2 equiv.), Py/AcOH, CH_2Cl_2; Glycosylation: TBSOTf, CH_2Cl_2, room temperature; 46 % (**20**).



Scheme 6. Reagents and conditions: a) TBSOTf, CH_3CN, room temperature, 78 % (**21**), 32 % (**22**). The ratio of α/β anomers was determined by integration of diagnostic peaks in the ^1H NMR spectra. Lev deprotection: $NH_2NH_2{\cdot}H_2O$ (2 equiv.), Py/AcOH, CH_2Cl_2; Glycosylation: TBSOTf, CH_2Cl_2, room temperature; 24 % (**23**); b) $(HF)_n$·Py, THF, 0 °C; $SO_3{\cdot}Me_3N$, DMF, 100–120 °C, MW heating, 25 %; c) $NH_2(CH_2)_2NH_2$, *n*-BuOH/CF_3CH_2OH, 120 °C, MW heating; NaOH, MeOH/CF_3CH_2OH; Ac_2O, Et_3N, MeOH, 78 %.

Table 1
IC_{50} values (μM) for compounds **1**, **2**, **10**, **11** and **25** obtained from fluorescence polarization competition assays using MK and FGF-2.

Compound	1	2	11	10	25
MK	32 ± 15[a]	9.9 ± 1.8[a]	4.2 ± 0.5[a]	0.23 ± 0.08[a]	254[b]
FGF-2	98 ± 34[a]	43 ± 18[a]	55 ± 15[a]	4.8 ± 0.5[a]	271[b]

[a] The IC_{50} and the error represent the average and the standard deviation from at least three independent experiments [b] Reported in reference 24.

values for the first one, also in good accordance with previously reported data [22–24].

Synthetic CS-E tetrasaccharide **25** (Fig. 2), containing the naturally occurring sequence of the polysaccharide, with GalN and GlcA units, displayed IC_{50} values of 254 and 271 μM for MK and FGF-2, respectively [24]. Similarly, K_D values in the millimolar/high micromolar range were recently reported for the interaction between CS-E tetra and hexasaccharides and MK [30]. Therefore, our results indicate that the synthesized oligosaccharides mimic CS-E natural sequences, binding to these growth factors even with higher affinity. This could be explained by the presence of the benzyl groups and the anomeric fluorinated tail, probably due to additional interactions established between these groups and the protein surface.

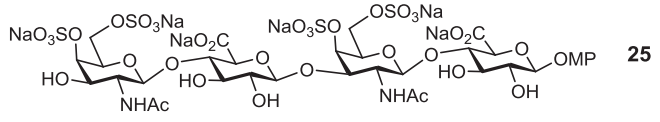


Fig. 2. Structure of CS-E tetrasaccharide **25**.

2.3. NMR studies

We carried out NMR experiments to study the 3D structure of tetrasaccharide **1** in D_2O. As mentioned before, compound **1** is a mimetic of an internal CS-E tetrasaccharide where the characteristic carboxylate at position 5 of the GlcA rings is not present and the GalN moiety is replaced by a GlcN unit. Thus, we performed an NMR analysis to check if its 3D structural features were similar to the standard CS-E oligosaccharides. Based on our experience, we focused on two key parameters: the ring conformation and the glycosidic linkage behaviour. Both sugar rings in CS, GalN and GlcA, used to be in a $^4C_1$ chair, and the coupling constants are usually used to assess that. However, the spectral overlapping hampered the precise quantification of couplings. Anyway, COSY-DQF cross peaks were compatible with the $^4C_1$ conformation, showing large coupling constants (7.5–10 Hz) for all the cases (Fig. 3 and Table 2).

Another way of assessing the ring conformation is by the analysis of the intraring H1-H3-H5 NOE contacts to discriminate if the ring conformation was compatible with the expected $^4C_1$ chair. The observed NOE cross peaks between H1 – H3, H3 – H5, and H1 – H5 indicated a mono-conformational behaviour, compatible with single $^4C_1$ conformation in all the rings (see Supplementary Material). On the other hand, the glycosidic dihedral angles analysis was compatible with a single *syn*-Ψ state, detected by the close contact between the anomeric and the ipso protons across the glycosidic linkages.

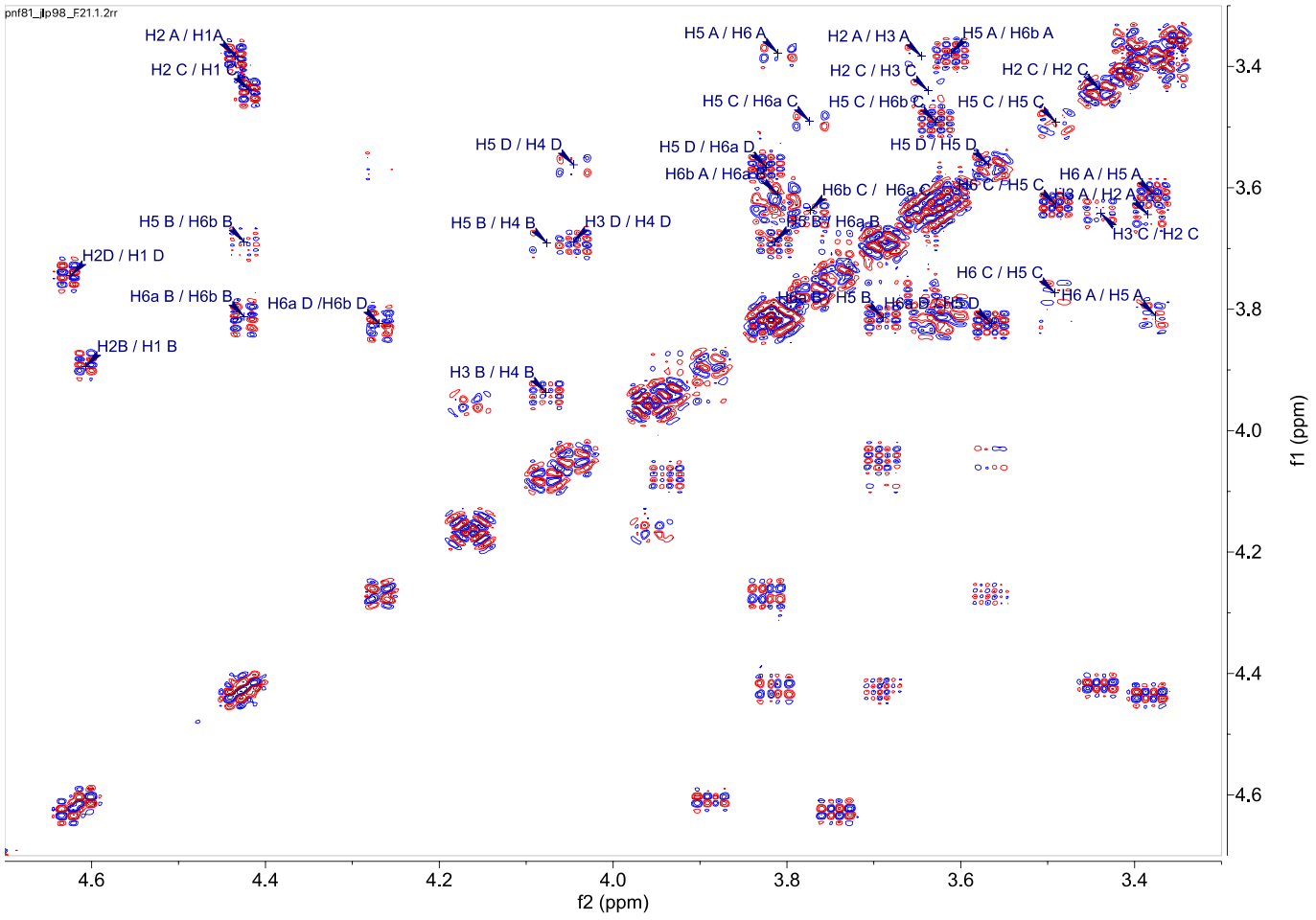


Fig. 3. COSY-DQF showing the assignment of the main cross-peaks.

Table 2
^{1}H Chemical shifts for **1**. Apparent $^{3}J_{HH}$ were calculated based on signal splitting.

	D		C		B		A	
	δ	$^{3}J_{HH}$	δ	$^{3}J_{HH}$	δ	$^{3}J_{HH}$	δ	$^{3}J_{HH}$
H-1	4.63	8.4	4.42	7.6	4.61	8.2	4.44	8.6
H-2	3.47	8.0	3.48	8.3	3.89	8.0	3.38	8.3
H-3	3.69	9.8	3.64		3.91	8.3	3.65	
H-4	4.05	8.9	3.77		4.08	8.8	3.81	
H-5	3.56	8.0	3.37		3.70		3.49	
H-6a	4.27		3.81		4.43		3.80	
H-6b	3.82	12.1	3.63		3.81		3.61	

Summarizing, the variations introduced by us in the standard GAGs structure, essentially the replacement of the carboxylate groups by hydroxymethyl moieties and the replacement of the GalN by GlcN rings, do not significantly affect the overall structure of tetrasaccharide **1**. Our NMR results qualitatively agree with the observations for natural CS oligosaccharides [33,41].

Then, we analysed the complex between compound **1** and MK in fast exchange conditions. For this, we prepared a 300 μL sample containing MK (20 μM) and tetrasaccharide **1** (1 mM) in 10 mM phosphate buffer with NaCl (150 mM). With this sample, we tried to perform the distances calculation analysis based on easy-ROESY experiments at 298.2 K using four or five tilted angles to calculate precise interprotonic distances and to characterize the conformational behaviour. Unfortunately, the calculation of accurate distance values was hampered by extensive signal overlapping. In any case, easy-ROESY experiments showed no evidences for different ring conformations or glycosidic linkage arrangements in the bound state. Therefore, the first conclusion from the binding NMR experiments was that the structure of the carbohydrate did not change upon binding. These results were similar to the previous ones with CS-E tetrasaccharides and mimetics [41], and they were also consistent with the behaviour of other GAGs.

To confirm the formation of the complex, we could not use the sign of the NOE cross-peaks because they already have a large molecule behaviour in the free state. For this reason, to be sure that the complex was formed, we performed solvent suppression STD experiments with 3 – 9 – 19 or W5 sequences (Fig. 4). The results indicated the binding between both biomolecules as peaks corresponding to the ligand **1** were present in the STD (Fig. 4). The analysis of the STD binding curves (Fig. 5 and Supplementary Material) did not reveal a preferential binding site in the tetrasacharide. This is compatible with the formation of a disordered complex that is relatively frequent in the case of interactions between short GAGs and lectins or growth factors [33,40,41].

2.4. Biological activity

Finally, we tested the capacity of tetrasaccharide **1** to modulate the function of MK in cellular cultures. Compound **10**, showing the highest binding affinity value for MK, was also included in the experiment. MK has a prominent role in neural regeneration [66]. In addition, its key role in tumor promotion and bad patient prognosis associated with high expression levels is now receiving increasing attention [67]. Initial works studying the transformation capacity of MK were performed in NIH3T3 cells [68,69], which show genomic imbalances highly

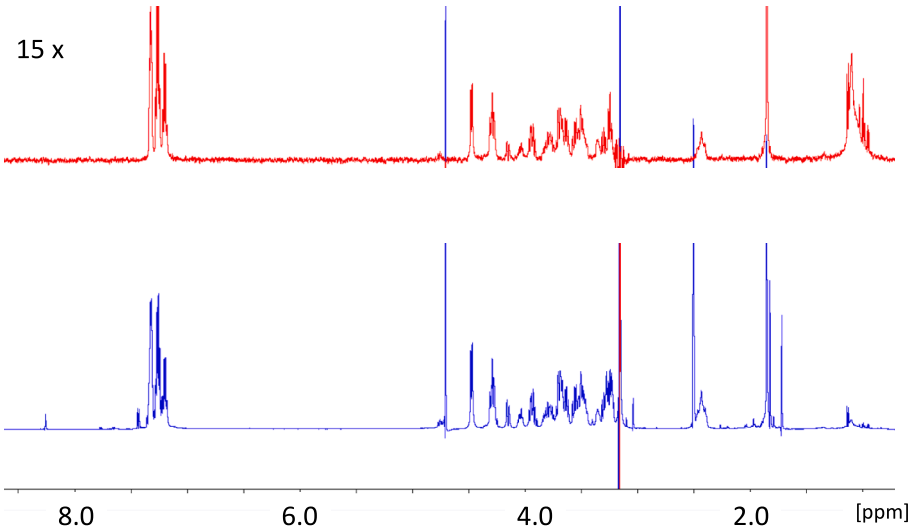


Fig. 4. STD experiment, with 3 s saturation time, solvent suppression with W5 sequence and spinlock (10 ms), reference at the bottom.

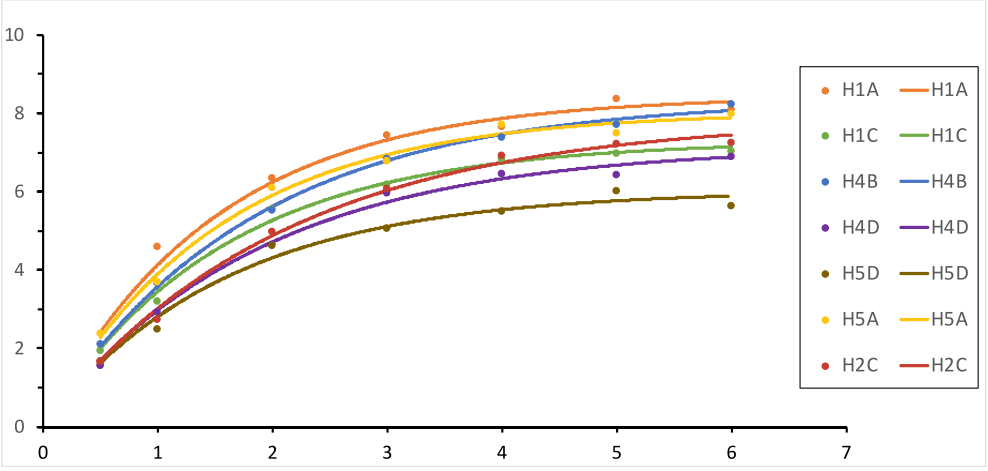


Fig. 5. STD growth rate. Experimental values (dots) fitted Box Lucas curves (lines).

resembling those found in human solid cancers of ectodermal origin [70]. Thus, this cell line constitutes an appropriate model for studying MK capacity to stimulate cell proliferation and transformation associated with cancer [41,68,69,71]. According to this, the addition of MK increased the growth rate in comparison to control conditions (Fig. 6). We then studied the effect of tetrasaccharides **1** and **10** on MK stimulating properties. Interestingly, as observed in Fig. 6, both tetrasaccharides **1** and **10** showed the ability of blocking MK-stimulating effect on NIH3T3 cells growth. Tetrasaccharide **10** was more efficient in doing that at day 4, which correlates with its higher affinity for MK. Importantly, we have previously observed that heparin displays a similar effect [41]. Thus, tetrasaccharides **1** and **10** seem to mimic at least some of the in vitro effects of this natural compound.

3. Conclusions

The use of GlcN and Glc building blocks, instead of GalN and GlcA units, facilitates the access to GAG-like sulfated oligosaccharides that retain their protein binding properties. Thus, the target oligosaccharides were efficiently assembled by an iterative 1 + 1 glycosylation approach using monosaccharidic trichloroacetimidates. First, we performed glycosylation reactions to install a perfluorinated tag at the anomeric position of the reducing end sugar. Our results highlight the importance of the acceptor nucleophilicity to explain the stereochemical outcome of the glycosylation reactions. The presence of the fluorinated tail allowed the straightforward isolation of reaction intermediates by F-SPE. Once the fully protected compounds were generated, the deprotection-sulfation steps afforded the desired GAG-like derivatives.

Fluorescence polarization assays demonstrated that the synthesized oligosaccharides bind to MK and FGF-2 in the micromolar range, displaying higher affinities than CS-E natural sequences, particularly in the case of MK. Our results suggest that the presence of the benzyl groups and the anomeric tail enhanced the interaction, in good agreement with our previous reports [23,41]. The binding between tetrasaccharide **1** and MK was also confirmed by STD-NMR. Moreover, NMR experiments indicated that the 3D structure of the mimetics was similar to that of the native oligomers, in terms of sugar ring and glycosidic linkage conformations.

MK induces growth in a variety of cell types, including cancer cells. Some of our synthetic compounds were able to block MK-stimulating effect on NIH3T3 cells proliferation, suggesting a potential therapeutic application for our derivatives.

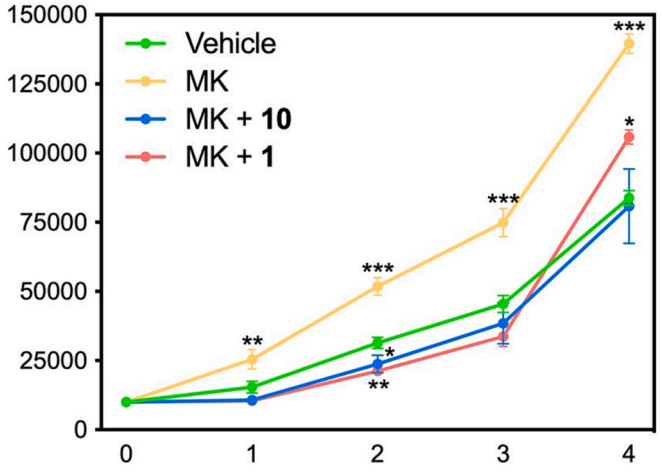


Fig. 6. Compounds **1** and **10** impair MK-mediated stimulation of cell growth. NIH3T3 cells were cultured in the absence (vehicle, control condition) or the presence of MK, with or without compounds **1**, or **10**. The number of cells was registered at days 0 to 4. The indicated time points correspond to means ± s.d. from three independent experiments. At each different day, statistical significance of differences between conditions was analyzed by one-way ANOVA ($P < 0.05$) followed by the Tukey's post-test. Statistical significance of differences between control (vehicle) and MK curves is indicated close to each time point (*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$).

4. Materials and Methods

4.1. Chemical synthesis

4.1.1. General synthetic procedures

Thin-layer chromatography (TLC) analyses were performed on silica gel 60F-254 precoated aluminium plates from Merck. The compounds were detected by UV visualization (λ = 254 nm) and by staining with 5 % v/v anisaldehyde-5 % v/v H_2SO_4-0.2 % v/v AcOH in EtOH or 0.2 % w/v cerium (IV) sufate-5 % w/v ammonium molybdate tetrahydrate in 1 M H_2SO_4 followed by heating at over 200 °C. Column chromatography was carried out on silica gel 60 (0.2–0.063 mm or 0.040–0.015 mm; Merck). F-SPE was performed using FluoroFlash silica gel from Sigma-Aldrich. Optical rotations were determined with a Perkin-Elmer 341 polarimeter, using a sodium lamp (λ = 589 nm) at 25 °C in 1 dm tubes. ^{1}H-, ^{19}F and ^{13}C NMR spectra were acquired on an Bruker 400 spectrometer. 1-D TOCSY, and 2-D COSY and HSQC experiments were routinely carried out to assist in signal assignment. Unit A refers to the reducing end monosaccharide in the NMR data. Electrospray mass spectra (ESI MS) were carried out with an Esquire 6000 ESI-Ion Trap or an amaZon SL mass spectrometer from Bruker. High resolution mass spectra (HR MS) were carried out by CITIUS (Universidad de Sevilla). Microwave-based reactions were performed using a Biotage Initiator Eight synthesizer in sealed reaction vessels.

General procedure for glycosidation with fluorinated alcohols. Monosaccharide trichloroacetimidate (1 equiv.) was coevaporated with toluene and dried under high vacuum. Then, the fluorinated alcohol (2 equiv.) was added and the mixture was dissolved in dry CH_3CN (1.5–2.5 mL) in the presence of activated 4 Å molecular sieves (MS, 150–225 mg). After stirring for 10 min at room temperature (rt), under an argon atmosphere, TBSOTf (0.3 equiv. for **3**, 0.07 equiv. for **4**, 0.2 equiv. for **17**) was added. After stirring for 30 min at rt, the reaction mixture was quenched with triethylamine (0.2 mL), filtered, and concentrated under reduced pressure. The crude material was purified by F-SPE to afford the corresponding glycosydation product.

General procedure for deprotection of Lev groups. The Lev-protected compound (0.142–0.035 mmol) was dissolved in CH_2Cl_2 (1.5 mL) and then treated with a 0.5 M solution of hydrazine monohydrate (2 equiv.) in Py/AcOH 3:2. After stirring at rt for 1.5 h, the reaction mixture was quenched with acetone (0.2 mL), diluted with CH_2Cl_2 and washed with 1 M HCl aqueous solution, saturated $NaHCO_3$ aqueous solution and H_2O. The organic layer was dried ($MgSO_4$), filtered and concentrated in vacuo. The crude product was purified by F-SPE to give the corresponding deprotected derivative.

General procedure for glycosidation. Glycosyl aceptor (1 equiv.) and glycosyl donor (2–3 equiv.) were coevaporated with toluene and dried under high vacuum. The mixture was dissolved in dry CH_2Cl_2 (1.5–2.0 mL) in the presence of activated 4 Å MS (150 mg approximately) and stirred, under an argon atmosphere, for 10 min. TBSOTf (7 mol% with respect to donor **4** or 20 mol% with respect to donors **3** and **17**) was then added. After stirring for 20–30 min at rt, the reaction mixture was quenched with triethylamine (0.3 mL), filtered, and concentrated under reduced pressure. The crude material was purified by F-SPE to afford the corresponding glycosydation product.

General procedure for F-SPE. As described previously [23], a glass chromatography column was packed with 5 g of FluoroFlash silica gel. Unless otherwise specified, precondition was performed with MeOH/H_2O 80:20 (15 mL), the crude sample (50–300 mg) was loaded on the column using DMF/H_2O 9:1 (1 mL) and the fluorophobic and fluorophilic elutions were carried out with 15 mL of MeOH/H_2O 80:20 and 20 mL of acetone, respectively. To regenerate the F-SPE column, we washed with additional acetone (20 mL).

1*H*,1*H*-perfluorononyl 2-*O*-benzoyl-3-*O*-benzyl-4-*O*-levulinoyl-6-*O*-pivaloyl-α,β-D-glucopyranoside (7): Following the general procedure, **3** (55 mg, 0.079 mmol) was glycosylated with 1*H*,1*H*-perfluoro-1-nonanol (80 mg, 0.177 mmol). The crude mixture was purified by F-SPE to give compound **7** as a 2.3:1 α/β mixture (43 mg, 55 %). TLC (toluene-EtOAc 8:1) Rf 0.34 and 0.21; Data for the α anomer: ^{1}H NMR (400 MHz, $CDCl_3$): δ 8.05–7.18 (m, 10*H*, Ar), 5.29 (d, 1H, $J_{1,2}$ = 3.8 Hz, H-1), 5.18 (m, 2H, H-2, H-4), 4.73 (2d, 2H, J = 11.6 Hz, CH_2(Bn)), 4.28–3.97 (m, 6H, H-3, H-5, H-6, $CH_2C_8F_{17}$), 2.75–2.46 (m, 4H, CH_2(Lev)), 2.18 (s, 3H, CH_3(Lev)), 1.25 (s, 9H, $C(CH_3)_3$); ^{13}C NMR (100 MHz, $CDCl_3$): δ 206.1, 178.0, 171.4, 165.7 (CO), 137.8–127.7 (Ar), 97.0 (C-1), 77.0 (C-3), 75.1 (CH_2(Bn)), 73.1 (C-2), 69.7 (C-4), 69.0 (C-5), 64.7 (t, $J_{C,F}$ = 26.5 Hz, $CH_2C_8F_{17}$), 61.9 (C-6), 38.9 ($C(CH_3)_3$), 37.8 (CH_2(Lev)), 29.7 (CH_3(Lev)), 27.8 (CH_2(Lev)), 27.1 ($C(CH_3)_3$); ^{19}F NMR (376 MHz, $CDCl_3$): δ −80.8 (t, 3F), −119.6 (m, 2F), −122.0 (m, 6F), −122.7 (m, 2F), −123.0 (m, 2F), −126.1 (m, 2F); Diagnostic peaks β anomer: ^{1}H NMR (400 MHz, $CDCl_3$): δ 5.35 (t, 0.43H, $J_{1,2}$ = $J_{2,3}$ = 8.4 Hz, H-2), 4.76 (d, 0.43H, H-1), 4.63 (s, 0.86H, CH_2(Bn)), 4.32 (dd, 0.43H, $J_{5,6}$ = 2.0 Hz, $J_{6,6}$ = 12.1 Hz, H-6), 3.75 (m, 0.43H, H-5); ^{13}C NMR (100 MHz, $CDCl_3$): δ 100.9 (C-1), 79.4 (C-3), 74.1 (CH_2(Bn)), 72.8, 72.6 (C-2, C-5), 69.8 (C-4); HR MS: *m/z*: calcd for $C_{39}H_{37}F_{17}O_{10}Na$: 1011.2007; found: 1011.1994 $[M + Na]^+$.

1*H*,1*H*,2*H*,2*H*-perfluorodecyl 2-*O*-benzoyl-3-*O*-benzyl-4-*O*-levulinoyl- 6-*O*-pivaloyl-α,β-D-glucopyranoside (8): Following the general procedure for glycosidation with fluorinated alcohols, **3** (94 mg, 0.134 mmol) was coupled with 1*H*,1*H*,2*H*,2*H*-perfluoro-1-decanol (124 mg, 0.268 mmol). The crude mixture was purified by F-SPE. The acetone fraction was concentrated to give **8** as a 1:8 α/β mixture (114 mg, 85 %). Standard silica gel chromatography column (toluene-EtOAc 12:1) afforded pure **8β** and **8α** derivatives as white amorphous solids.

8β: TLC (toluene-EtOAc 5:1) Rf 0.37; $[\alpha]_D^{20}$ + 15° (*c* 1.0, $CHCl_3$); ^{1}H NMR (400 MHz, $CDCl_3$): δ 8.02–7.15 (m, 10*H*, Ar), 5.32 (dd, 1H, $J_{2,3}$ = 9.4 Hz, $J_{1,2}$ = 7.9 Hz, H-2), 5.18 (t, 1H, $J_{3,4}$ = $J_{4,5}$ = 9.6 Hz, H-4), 4.63 (m, 3H, CH_2(Bn), H-1), 4.31 (dd, 1H, $J_{5,6}$ = 2.2 Hz, $J_{6,6}$ = 12.2 Hz, H-6), 4.14 (m, 2H, H-6, $CH_2CH_2C_8F_{17}$), 3.91 (t, 1H, H-3), 3.81 (m, 1H, $CH_2CH_2C_8F_{17}$), 3.74 (m, 1H, H-5), 2.76–2.33 (m, 6H, CH_2(Lev), $CH_2CH_2C_8F_{17}$), 2.18 (s, 3H, CH_3(Lev)), 1.24 (s, 9H, $C(CH_3)_3$); ^{13}C NMR (100 MHz, $CDCl_3$): δ 206.1, 178.1, 171.4, 165.0 (CO), 137.5–127.7 (Ar), 101.1(C-1), 79.7 (C-3), 74.0 (CH_2(Bn)), 73.0 (C-2), 72.6 (C-5), 70.1 (C-4), 62.1 (C-6), 61.5 ($CH_2CH_2C_8F_{17}$), 38.8 ($C(CH_3)_3$), 37.8 (CH_2(Lev)), 31.5 (t, $J_{C,F}$ = 21.4 Hz, $CH_2CH_2C_8F_{17}$), 29.7 (CH_3(Lev)), 27.9 (CH_2(Lev)), 27.0 ($C(CH_3)_3$); ^{19}F NMR (376 MHz, $CDCl_3$): δ −80.8 (t, 3F), −113.3 (m, 2F), −122.0 (m, 6F), −122.7 (m, 2F), −123.6 (m, 2F),

−126.1 (m, 2F); HR MS: *m/z*: calcd for $C_{40}H_{39}F_{17}O_{10}Na$: 1025.2164; found: 1025.2152 $[M + Na]^+$.

8α: TLC (toluene-EtOAc 5:1) Rf 0.50; $[\alpha]_D^{20}$ + 43° (*c* 0.8, $CHCl_3$); ^1H NMR (400 MHz, $CDCl_3$): δ 8.06–7.21 (m, 10*H*, Ar), 5.23 (d, 1H, $J_{1,2}$ = 3.7 Hz, H-1), 5.13 (m, 2H, H-2, H-4), 4.73 (2d, 2H, *J* = 11.6 Hz, CH_2(Bn)), 4.21–4.15 (m, 3H, H-3, H-6), 4.03 (m, 2H, H-5, $CH_2CH_2C_8F_{17}$), 3.72 (m, 1H, $CH_2CH_2C_8F_{17}$), 2.73–2.42 (m, 6H, CH_2(Lev), $CH_2CH_2C_8F_{17}$), 2.18 (s, 3H, CH_3(Lev)), 1.25 (s, 9H, $C(CH_3)_3$); ^{13}C NMR (100 MHz, $CDCl_3$, selected data from HSQC experiment): δ 96.1 (C-1), 77.0 (C-3), 75.0 (CH_2(Bn)), 73.0 (C-2), 69.8 (C-4), 68.3 (C-5), 61.7 (C-6), 60.1 ($CH_2CH_2C_8F_{17}$); ^{19}F NMR (376 MHz, $CDCl_3$): δ −80.7 (t, 3F), −113.2 (m, 2F), −121.8 (m, 6F), −122.7 (m, 2F), −123.5 (m, 2F), −126.0 (m, 2F); HR MS: *m/z*: calcd for $C_{40}H_{39}F_{17}O_{10}Na$: 1025.2164; found: 1025.2155 $[M + Na]^+$.

1*H*,1*H*,2*H*,2*H*-perfluorodecyl *O*-(2-deoxy-3-*O*-levulinoyl-2-phthalimido-4,6-*O*-di-*tert*-butylsilylene-β-D-glucopyranosyl)-(1 → 4)-*O*-(2-*O*-benzoyl-3-*O*-benzyl-6-*O*-pivaloyl-β-D-glucopyranosyl)-(1 → 3)-*O*-(2-deoxy-2-phthalimido-4,6-*O*-di-*tert*-butylsilylene-β-D-glucopyranosyl)-(1 → 4)-2-*O*-benzoyl-3-*O*-benzyl-6-*O*-pivaloyl-β-D-glucopyranoside (9): Following the general procedure for deprotection of Lev groups, compound **8β** (90 mg, 0.090 mmol) was transformed into the corresponding Lev-deprotected derivative (70 mg). ESI MS: *m/z*: calcd for $C_{35}H_{33}F_{17}O_8Na$: 927.2; found: 927.1 $[M + Na]^+$.

This acceptor monosaccharide (70 mg, 0.077 mmol) was glycosylated with donor **4** (107 mg, 0.155 mmol), following the general procedure for glycosidation, to afford the corresponding disaccharide (110 mg). ESI MS: *m/z*: calcd for $C_{62}H_{68}F_{17}NO_{16}SiNa$: 1456.4; found: 1456.4 $[M + Na]^+$.

This disaccharide (110 mg, 0.077 mmol) was submitted to the general procedure for Lev deprotection to yield the corresponding 3-OH disaccharide (91 mg). ESI MS: *m/z*: calcd for $C_{57}H_{62}F_{17}NO_{14}SiNa$: 1358.4; found: 1358.4 $[M + Na]^+$.

This disaccharide acceptor (91 mg, 0.068 mmol) was glycosylated with donor **3** (134 mg, 0.191 mmol), as described in the general procedure, to give the corresponding trisaccharide (125 mg). ESI MS: *m/z*: calcd for $C_{87}H_{96}F_{17}NO_{23}SiNa$: 1896.6; found: 1896.6 $[M + Na]^+$.

As described in the general procedure, this trisaccharide (124 mg, 0.066 mmol) was deprotected to afford the corresponding delevulinoylated trisaccharide (104 mg). ESI MS: *m/z*: calcd for $C_{82}H_{90}F_{17}NO_{21}SiNa$: 1798.5; found: 1798.5 $[M + Na]^+$.

Finally, the glycosyl acceptor trisaccharide (101 mg, 0.057 mmol) was treated with donor **4** (110 mg, 0.150 mmol), following the general procedure for glycosidation. After further purification by silica gel column chromatography (toluene-EtOAc 11:1), pure tetrasaccharide **9** was obtained as a white amorphous solid (100 mg, 48 % from **8β**, 6 steps). TLC (toluene-EtOAc 10:1) Rf 0.23; $[\alpha]_D^{20}$ −17° (*c* 1, $CHCl_3$); ^1H NMR (400 MHz, $CDCl_3$): δ 7.92–7.35 (m, 18H, Ar), 7.14–7.02 (m, 10*H*, Ar), 5.63 (dd, 1H, $J_{2,3}$ = 10.6 Hz, $J_{3,4}$ = 8.9 Hz, H-3D), 5.36 (d, 1H, $J_{1,2}$ = 8.2 Hz, H-1D), 5.13–5.05 (m, 2H, H-2A, H-1B), 4.92 (t, 1H, $J_{1,2} = J_{2,3}$ = 7.8 Hz, H-2C), 4.74–4.57 (m, 4H, 3 × CH_2(Bn), H-1C), 4.44 (dd, 1H, $J_{2,3}$ = 10.5 Hz, $J_{3,4}$ = 8.4 Hz, H-3B), 4.36 (m, 2H, CH_2(Bn), H-1A), 4.30 (dd, 1H, $J_{5,6}$ = 2.4 Hz, $J_{6,6}$ = 12.0 Hz, H-6A or C), 4.20–4.12 (m, 3H, H-6A or C, H-2B, H-2D), 3.96 (dd, 1H, $J_{5,6}$ = 4.8 Hz, $J_{6,6}$ = 10.5 Hz, H-6B), 3.91–3.83 (m, 5H, H-4A, H-4B, H-4C, H-6D, $CH_2CH_2C_8F_{17}$), 3.71 (t, 1H, $J_{4,5}$ = 9.2 Hz, H-4D), 3.63–3.57 (m, 3H, H–3A, H-6A or C, $CH_2CH_2C_8F_{17}$), 3.55–3.50 (m, 3H, H-6B, H-3C, H-6A or C), 3.47–3.34 (m, 2H, H-5D, H-5B), 3.31–3.24 (m, 3H, H-6D, H-5C, H-5A), 2.58–2.38 (m, 4H, CH_2(Lev)), 2.26 (m, 2H, $CH_2CH_2C_8F_{17}$), 1.95 (s, 3H, CH_3(Lev)), 1.20–1.13 (2 s, 2 × 9H, 2 × $C(CH_3)_3$), 0.98–0.93 (4 s, 36H, $Si(C(CH_3)_3)_2$); ^{13}C NMR (100 MHz, $CDCl_3$): δ 205.6, 177.3, 177.2, 171.9, 168.1, 167.7, 165.1, 164.8 (CO), 138.1–123.6 (Ar), 100.8 (C-1A), 99.8 (C-1C), 97.7, 97.6 (C-1B, C-1D), 79.8 (C-3C), 79.5 (C-3A), 77.6 (C-3B), 77.1, 75.9, 75.7, 75.5 (C-4A, C-4B, C-4C, C-4D), 74.4, 74.0 (2 × CH_2(Bn), C-2C), 73.7, 73.3 (C-5C, C-5A), 72.6, 72.4 (C-3D, C-2A), 70.5, 70.3 (C–5B, C–5D), 65.9, 65.6 (C-6B, C–6D), 62.6, 61.8 (C-6A, C-6C), 61.2 ($CH_2CH_2C_8F_{17}$), 55.8, 55.1 (C-2B, C-2D), 38.7 ($C(CH_3)_3$), 37.8 (CH_2(Lev)), 31.5 (t, $J_{C,F}$ = 21.3 Hz, $CH_2CH_2C_8F_{17}$), 29.4 (CH_3(Lev)), 27.8 (CH_2(Lev)), 27.3–26.8 ($C(CH_3)_3$, $Si(C(CH_3)_3)_2$), 22.5, 22.4, 19.9, 19.8 ($Si(C(CH_3)_3)_2$); ^{19}F NMR (376 MHz, $CDCl_3$): δ −80.8 (t, 3F), −113.4 (m, 2F), −122.0 (m, 6F), −122.8 (m, 2F), −123.7 (m, 2F), −126.1 (m, 2F); HR MS: *m/z*: calcd for $C_{109}H_{125}F_{17}O_{29}N_2Si_2Na$: 2327.7527; found: 2327.7497 $[M + Na]^+$.

1*H*,1*H*,2*H*,2*H*-perfluorodecyl *O*-(2-deoxy-3-*O*-levulinoyl-2-phthalimido-4,6-di-*O*-sulfo-β-D-glucopyranosyl)-(1 → 4)-*O*-(2-*O*-benzoyl-3-*O*-benzyl-6-*O*-pivaloyl-β-D-glucopyranosyl)-(1 → 3)-*O*-(2-deoxy-2-phthalimido-4,6-di-*O*-sulfo-β-D-glucopyranosyl)-(1 → 4)-2-*O*-benzoyl-3-*O*-benzyl-6-*O*-pivaloyl-β-D-glucopyranoside (10): An excess of $(HF)_n$·Py (0.1 mL, 3.9 mmol) was added at 0 °C under an argon atmosphere to a solution of **9** (45 mg, 19.5 μmol) in dry THF (1.5 mL). After 6 h at 0 °C, the mixture was diluted with CH_2Cl_2 and washed with H_2O, saturated $NaHCO_3$ solution and H_2O. The organic layers were dried ($MgSO_4$), filtered and concentrated in vacuo to give the corresponding tetraol (40 mg) as a white amorphous solid that was directly submitted to the sulfation reaction. TLC (toluene-EtOAc 1:2) Rf 0.2; ESI MS: *m/z*: calcd for $C_{93}H_{93}F_{17}N_2O_{29}Na$: 2048.6; found: 2048.3 $[M + Na]^+$.

The tetraol (19.5 μmol) and sulfur trioxide–trimethylamine complex (108 mg, 0.78 mmol) were dissolved in dry DMF (2.5 mL) and heated at 100 °C for 30 min using microwave radiation (30 W average power). The reaction vessel was cooled and Et_3N (300 μL) was added. The crude solution was purified by Sephadex LH 20 chromatography (CH_2Cl_2-MeOH 1:1) and silica gel column chromatography (EtOAc-MeOH-H_2O 32:5:3 → EtOAc-MeOH-H_2O 24:5:3). Finally, the residue was eluted from a Dowex 50WX2-Na^+ column (MeOH) to obtain **10** as sodium salt (34 mg, 72 %, 2 steps, white amorphous solid). TLC (EtOAc-MeOH-H_2O 24:5:3) Rf 0.22; ^1H NMR (400 MHz, CD_3OD): δ 7.97–7.39 (m, 18H, Ar), 7.07–6.99 (m, 10*H*, Ar), 5.86 (dd, 1H, $J_{2,3}$ = 10.8 Hz, $J_{3,4}$ = 8.8 Hz, H-3D), 5.48 (d, 1H, $J_{1,2}$ = 8.3 Hz, H-1D), 5.31 (d, 1H, $J_{1,2}$ = 8.3 Hz, H-1B), 5.01 (t, 1H, $J_{1,2} = J_{2,3}$ = 8.5 Hz, H-2A or C), 4.94 (dd, 1H, $J_{1,2}$ = 7.8 Hz, $J_{2,3}$ = 8.7 Hz, H-2A or C), 4.85–4.76 (m, 4H, 2 × CH_2(Bn), H-3B, H-6B), 4.70–4.65 (2 × d, 2H, H-1A, H-1C), 4.60–4.52 (m, 3H, 2 × CH_2(Bn), H-6D), 4.39–4.28 (m, 4H, 2 × H-6A or C, H-4B, H-4D), 4.22–4.14 (m, 3H, H-2B, H-2D, H-6B), 4.09–3.89 (m, 7H, H-6D, H-5B, H-4A, H-4C, H-3 A or C, H-6A or C, $CH_2CH_2C_8F_{17}$), 3.86–3.74 (m, 4H, H–5A or C, H-6A or C, H-5D, $CH_2CH_2C_8F_{17}$), 3.59 (t, 1H, $J_{3,4}$ = 8.6 Hz, H-3A or C), 3.49 (m, 1H, H-5A or C), 2.61–2.28 (m, 6H, CH_2(Lev), $CH_2CH_2C_8F_{17}$), 1.90 (s, 3H, CH_3(Lev)), 1.24–1.13 (2 s, 2 × 9H, 2 × $C(CH_3)_3$); ^{13}C NMR (100 MHz, CD_3OD): δ 207.4, 178.4, 178.0, 172.5, 165.8, 165.6 (CO), 138.0–123.2 (Ar), 100.3, 99.2 (C-1A, C-1C), 96.9 (C-1D), 95.4 (C-1B), 79.8 (C-3A or C), 76.4 (C-4B), 77.4, 75.7, 75.2 (4C: C-3B, C-4A, C-4C, C-3A or C), 74.6 (C-4D), 73.7–72.4 (8C: 2 × CH_2(Bn), C-2A, C-2C, C-5A, C-5B, C-5C, C-5D), 70.5 (C-3D), 67.4, 66.6 (C-6B, C–6D), 62.6, 62.4 (C-6A, C-6C), 61.1 ($CH_2CH_2C_8F_{17}$), 56.0, 55.1 (C-2B, C-2D), 38.5, 38.4 ($C(CH_3)_3$), 37.1 (CH_2(Lev)), 30.9 (t, $J_{C,F}$ = 18.7 Hz, $CH_2CH_2C_8F_{17}$), 27.84, 27.75 (CH_3(Lev), CH_2(Lev)), 26.4, 26.2 ($C(CH_3)_3$); ^{19}F NMR (376 MHz, CD_3OD): δ −82.4 (t, 3F), −114.3 (m, 2F), −123.0 (m, 6F), −123.7 (m, 2F), −124.6 (m, 2F), −127.3 (m, 2F); HR MS: *m/z*: calcd for $C_{93}H_{89}F_{17}O_{41}N_2S_4Na_2$: 1193.1679; found: 1193.1730 $[M-2Na]^{2-}$.

1*H*,1*H*,2*H*,2*H*-perfluorodecyl *O*-(2-acetamido-2-deoxy-4,6-di-*O*-sulfo-β-D-glucopyranosyl)-(1 → 4)-*O*-(3-*O*-benzyl-β-D-glucopyranosyl)-(1 → 3)-*O*-(2-acetamido-2-deoxy-4,6-di-*O*-sulfo-β-D-glucopyranosyl)-(1 → 4)-3-*O*-benzyl-β-D-glucopyranoside (1): Ethylene diamine (107 μL, 1.60 mmol) was added, under an argon atmosphere, to a solution of **10** (13 mg, 5.3 μmol) in a mixture of *n*-BuOH (1.5 mL) and 2,2,2-trifluoroethanol (0.1 mL). Then, the reaction mixture was subjected to microwave irradiation (38 W average power) for 90 min at 120 °C (3 cycles, 30 min each). The reaction vessel was cooled, and the mixture was concentrated to dryness. The residue was disolved in MeOH/CF_3CH_2OH (2.5 mL/0.1 mL) and an aqueous solution of NaOH (4 M, 270 μL) was added. After stirring for 48 h at room temperature, the reaction mixture was neutralized with Amberlite IR-120 (H^+) resin, filtered, and concentrated to give the desired amine intermediate. Triethylamine (20 μL, 0.14 mmol) and acetic anhydride (20 μL, 0.21

mmol) were added to a cooled (0 °C) solution of this amine derivative in MeOH (1.5 mL). After stirring for 3 h at room temperature, additional triethylamine (20 μL, 0.14 mmol) and acetic anhydride (20 μL, 0.21 mmol) were added at 0 °C to complete the reaction. After further stirring for 2 h at room temperature, Et_3N (0.2 mL) was added and the mixture was concentrated to dryness. The residue was then purified by silica gel column chromatography (EtOAc-MeOH-H_2O 24:5:3 → EtOAc-MeOH-H_2O 20:5:3) and finally eluted from a Dowex 50WX2-Na^+ column (MeOH) to obtain **1** as sodium salt (7.4 mg, 78 %, white amorphous solid). TLC (EtOAc-MeOH-H_2O 16:5:3) Rf 0.38; ^{1}H NMR (400 MHz, CD_3OD): δ 7.53–7.24 (m, 10*H*, Ar), 5.04–4.93 (m, 4H, CH_2(Bn)), 4.80 (d, 1H, $J_{1,2}$ = 7.6 Hz, H-1B or D), 4.72 (d, 1H, $J_{1,2}$ = 8.1 Hz, H-1B or D), 4.66 (dd, 1H, $J_{5,6}$ = 2.7 Hz, $J_{6,6}$ = 11.5 Hz, H-6B or D), 4.52 (d, 1H, $J_{1,2}$ = 7.7 Hz, H-1A or C), 4.45 (dd, 1H, $J_{5,6}$ = 2.2 Hz, $J_{6,6}$ = 11.3 Hz, H-6B or D), 4.40 (d, 1H, $J_{1,2}$ = 7.8 Hz, H-1A or C), 4.24–4.18 (m, 2H, H-4B or D, $CH_2CH_2C_8F_{17}$), 4.11–3.85 (m, 10*H*, H-4A, H-4C, H-4B or D, 2 × H-6A or C, 2 × H-6B or D, H-2B or D, H-3B or D, $CH_2CH_2C_8F_{17}$), 3.81–3.68 (m, 7H, H-3A, H-3C, H-2B or D, H-3B or D, H-5B or D, 2 × H-6A or C), 3.64 (m, 1H, H-5A or C), 3.57–3.51 (m, 3H, H-5A or C, H-2A or C, H-5B or D), 3.42 (dd, 1H, $J_{2,3}$ = 8.8 Hz, H-2A or C), 2.66–2.53 (m, 2H, $CH_2CH_2C_8F_{17}$), 2.01–2.00 (2 × s, 6H, NHAc); ^{13}C NMR (100 MHz, CD_3OD, selected data from HSQC experiment): δ 103.2, 102.8 (C-1A, C-1C), 99.0, 98.6 (C-1B, C-1D), 81.3, 80.2 (C-3A, C-3C), 78.6 (C-3B or D), 76.8, 75.3, 74.5 (C-4A, C-4B, C-4C, C-4D), 75.1, 74.9, 73.6, 73.2, 72.6 (C-5A, C-5B, C-5C, C-5D, C-2A, C-2C, C-3B or D), 72.7, 71.8 (2 × CH_2(Bn)), 67.4, 67.1 (C-6B, C–6D), 61.1 ($CH_2CH_2C_8F_{17}$), 60.2 (C-6A, C-6C), 55.7, 55.3 (C-2B, C-2D), 30.7 ($CH_2CH_2C_8F_{17}$), 20.7 (NHAc); ^{19}F NMR (376 MHz, CD_3OD): δ −82.3 (t, 3F), −114.4 (m, 2F), −122.8 (m, 6F), −123.7 (m, 2F), −124.6 (m, 2F), −127.2 (m, 2F); HR MS: *m/z*: calcd for $C_{52}H_{59}F_{17}O_{33}N_2S_4Na_3$: 1759.1310; found: 1759.1278 [*M*−Na]$^-$.

1*H*,1*H*,2*H*,2*H*-perfluorodecyl *O*-(2-benzamido-2-deoxy-4,6-di-*O*-sulfo-β-D-glucopyranosyl)-(1 → 4)-*O*-(3-*O*-benzyl-β-D-glucopyranosyl)-(1 → 3)-*O*-(2-benzamido-2-deoxy-4,6-di-*O*-sulfo-β-D-glucopyranosyl)-(1 → 4)-3-*O*-benzyl-β-D-glucopyranoside (11): As described for the preparation of **1**, **10** (27 mg, 11 μmol) was treated with ethylendiamine and then NaOH to deprotect acyl (Bz, Lev and Piv) and *N*-phthaloyl groups. The diamine intermediate was dissolved in MeOH (2.5 mL), and then triethylamine (40 μL, 0.29 mmol) and benzoic anhydride (100 mg, 0.44 mmol) were added at 0 °C. After stirring for 3 h at room temperature, additional triethylamine (40 μL, 0.29 mmol) and benzoic anhydride (100 mg, 0.44 mmol) were added. After further stirring for 3 h at room temperature, Et_3N (0.3 mL) was added and the mixture was concentrated to dryness. The residue was then purified by silica gel column chromatography (EtOAc-MeOH-H_2O 28:5:3 → EtOAc-MeOH-H_2O 20:5:3) and finally eluted from a Dowex 50WX2-Na^+ column (MeOH) to obtain **11** as sodium salt (16.3 mg, 78 %, white amorphous solid). TLC (EtOAc-MeOH-H_2O 16:5:3) Rf 0.45; ^{1}H NMR (400 MHz, CD_3OD): δ 7.90–7.21 (m, 20H, Ar), 5.03–4.78 (m, 6H, CH_2(Bn), H-1B, H-1D), 4.68 (br d, 1H, $J_{6,6}$ = 11.1 Hz, H-6B or D), 4.55 (d, 1H, $J_{1,2}$ = 7.4 Hz, H-1A or C), 4.49 (br d, 1H, $J_{6,6}$ = 10.7 Hz, H-6B or D), 4.38 (d, 1H, $J_{1,2}$ = 7.8 Hz, H-1A or C), 4.31–4.01 (m, 8H, H-4B, H-4D, $CH_2CH_2C_8F_{17}$, H-2B, H-2D, H-3B or D, H-6B, H-6D), 3.94–3.80 (m, 7H, H-4A, H-4C, $CH_2CH_2C_8F_{17}$, H-3B or D, H-5B or D, H-6A, H-6C), 3.75–3.48 (m, 8H, H-5A, H-5C, H-6A, H-6C, H-2A or C, H-3A, H-3C, H-5B or D), 3.41 (m, 1H, H-2A or C), 2.57 (m, 2H, $CH_2CH_2C_8F_{17}$); ^{13}C NMR (100 MHz, CD_3OD, selected data from HSQC experiment): δ 102.7 (C-1A, C-1C), 98.7 (C-1B, C-1D), 81.5, 80.9 (C-3A, C-3C), 77.9, 77.0 (C-3B or D, C-4B or D), 75.0–72.5 (C-3B or D, C-4B or D, C-4A, C-4C, C-5A, C-5B, C-5C, C-5D, C-2A, C-2C), 72.5, 71.9 (2 × CH_2(Bn)), 67.5, 67.1 (C-6B, C–6D), 61.2 ($CH_2CH_2C_8F_{17}$), 60.4, 60.1 (C-6A, C-6C), 56.2, 55.8 (C-2B, C-2D), 30.8 ($CH_2CH_2C_8F_{17}$); ^{19}F NMR (376 MHz, CD_3OD): δ −82.3 (t, 3F), −114.4 (m, 2F), −122.8 (m, 6F), −123.7 (m, 2F), −124.6 (m, 2F), −127.2 (m, 2F); HR MS: *m/z*: calcd for $C_{62}H_{64}F_{17}O_{33}N_2S_4Na_4$: 1907.1588; found: 1907.1570 [*M* + H]$^+$.

4-Methoxyphenyl 3,4,6-tri-*O*-acetyl-2-deoxy-2-trifluoroacetamido-β-D-glucopyranoside (13): TMSOTf (21 μL, 0.114 mmol) was added to a cooled (0 °C) solution of **12** (336 mg, 0.76 mmol) and 4-methoxyphenol (188 mg, 1.52 mmol) in dry CH_2Cl_2 (4 mL) in the presence of activated 4 Å MS (400 mg). The mixture was stirred for 2 h at 0 °C and Et_3N (0.3 mL) was then added. The crude mixture was filtered and concentrated under reduced pressure. The residue was purified by silica gel chromatography (toluene/EtOAc 3:1) to give known **13** as a white amporphous solid (304 mg, 79 %). TLC (3:1 toluene/EtOAc) Rf 0.25; ^{1}H NMR (400 MHz, $CDCl_3$): δ 6.97–6.81 (m, 5H, NH, Ar), 5.39 (t, 1H, $J_{2,3}$ = $J_{3,4}$ = 10.0 Hz, H-3), 5.17 (t, 1H, H-4), 5.08 (d, 1H, $J_{1,2}$ = 8.3 Hz, H-1), 4.34–4.19 (m, 3H, H-2, H-6), 3.87 (m, 1H, H-5), 3.79 (s, 3H, Me (OMP)), 2.11, 2.07, 2.04 (3 s, 9H, CH_3(Ac)).

4-Methoxyphenyl 2-deoxy-2-trifluoroacetamido-β-D-glucopyranoside (14): Compound **13** (2.64 g, 5.19 mmol) was dissolved in MeOH (45 mL) and NaOMe (359 μL, 2.17 M solution in MeOH) was added. After 2.5 h, more MeOH (55 mL) was added and the reaction mixture was neutralized with Amberlite acidic resin, filtered and concentrated to give known **14** as a white amorphous solid (1.94 g, 98 %). TLC (8:1 CH_2Cl_2 /MeOH) R_f 0.28; ^{1}H NMR (400 MHz, CD_3OD): δ 6.99–6.84 (m, 4H, Ar), 4.97 (d, 1H, $J_{1,2}$ = 8.4 Hz, H-1), 3.94 (m, 2H, H-2, H-6), 3.76 (m, 4H, H-6, Me (OMP)), 3.64 (t, 1H, $J_{2,3}$ = $J_{3,4}$ = 9.2 Hz, H-3), 3.45 (m, 2H, H-4, H-5).

4-Methoxyphenyl 2-deoxy-4,6-*O*-di-*tert*-butylsilylene-2-trifluoroacetamido-β-D-glucopyranoside (15): Compound **14** (1.94 g, 5.10 mmol) was dissolved in dry Py (64 mL) and cooled (0 °C). Di-*tert*-butylsilyl bis(trifluoromethanesulfonate) ($tBu_2Si(OTf)_2$, 1.82 mL, 5.61 mmol) was added and the mixture was stirred at room temperature for 1 h. TLC analysis showed unreacted starting material and therefore additional $tBu_2Si(OTf)_2$ (1.00 mL, 3.08 mmol) was added at 0 °C. After stirring for 1 h at rt, the reaction mixture was quenched with MeOH (1 mL), diluted with CH_2Cl_2, and washed with 2 N HCl, saturated aqueous $NaHCO_3$, and water. The organic phase was dried ($MgSO_4$), filtered and concentrated to dryness. The residue was purified by column chromatography (toluene-EtOAc 10:1) to afford **15** as a white amorphous solid (1.87 g, 70 %). TLC (toluene-EtOAc 10:1) Rf 0.23; $[\alpha]_D^{20}$ −25° (*c* 1.0, $CHCl_3$); ^{1}H NMR (400 MHz, $CDCl_3$): δ 6.94–6.81 (m, 5H, NH, Ar), 5.32 (d, 1H, $J_{1,2}$ = 8.4 Hz, H-1), 4.23 (dd, 1H, $J_{5,6a}$ = 5.2 Hz, $J_{6a,6b}$ = 10.2 Hz, H-6a), 4.08 (br t, 1H, H-3), 3.98 (t, 1H, $J_{5,6b}$ = 10.2 Hz, H-6b), 3.78 (m, 5H, H-2, H-4, Me (OMP)), 3.56 (m, 1H, H-5), 2.94 (d, 1H, $J_{3,OH}$ = 2.0 Hz, OH), 1.09–1.01 (2 s, 18H, Si(C(CH_3)$_3$)$_2$); ^{13}C NMR (100 MHz, $CDCl_3$): δ 157.6 (q, $^2J_{C,F}$ = 37.1 Hz, $COCF_3$), 155.9–114.7 (Ar), 115.7 (q, $^1J_{C,F}$ = 288.9 Hz, $COCF_3$), 99.4 (C-1), 77.6 (C-4), 72.6 (C-3), 70.4 (C-5), 66.0 (C-6), 57.4, 55.6 (C-2, Me (OMP)), 27.4, 26.9 (Si(C(CH_3)$_3$)$_2$), 22.7, 19.9 (Si($C(CH_3)_3$)$_2$); ^{19}F NMR (376 MHz, $CDCl_3$): δ −75.8 (s, 3F); HR MS: *m/z*: calcd for $C_{23}H_{34}F_3O_7NSiNa$: 544.1949; found: 544.1942 [*M* + Na]$^+$.

2-deoxy-3-*O*-levulinoyl-4,6-*O*-di-*tert*-butylsilylene-2-trifluoroacetamido-α,β-D-glucopyranose (16): LevOH (2.2 mL, 21.5 mmol) was added at 0 °C to a solution of 1,3-dicyclohexylcarbodiimide (2.22 g, 10.8 mmol) in CH_2Cl_2 (30 mL). After stirring for 5 min at room temperature, the mixture was cooled and filtered, and the urea precipitate was washed with additional CH_2Cl_2 (15 mL). The resulting Lev_2O solution (10.8 mmol) was added at room temperature to a mixture of **15** (1.87 g, 3.59 mmol) and DMAP (66 mg, 0.54 mmol). The mixture was stirred for 2.5 h, diluted with CH_2Cl_2, and washed with saturated aqueous $NaHCO_3$, and H_2O. The organic phase was dried ($MgSO_4$), filtered and concentrated to dryness. The residue was purified by column chromatography (hexane-EtOAc 5:1) to give the 3-*O*-levulinated monosaccharide (2.16 g, 97 %).

CAN (12 mL of a 1.16 M solution in H_2O) was added at 0 °C to a solution of the levulinated compound (2.16 g, 3.48 mmol) in CH_2Cl_2/MeCN (1:2; 108 mL). After stirring for 1.5 h at 0 °C, the reaction mixture was diluted with EtOAc, washed with H_2O, saturated aqueous $NaHCO_3$, and brine. The organic phase was dried ($MgSO_4$), filtered and concentrated to dryness. The residue was purified by column chromatography (hexane-EtOAc 4:1) to afford **16** as a 9:1 α/β mixture (1.51 mg, 84 %, white amorphous solid). TLC (hexane-EtOAc 4:1) Rf 0.21. Data for α anomer: ^{1}H NMR (400 MHz, $CDCl_3$): δ 6.90 (d, 1H, $J_{2,NH}$ = 9.2 Hz, NH),

5.24 (m, 2H, H-1, H-3), 4.38 (br s, 1H, OH), 4.21 (ddd, 1H, $J_{1,2}$ = 3.6 Hz, H-2), 4.09 (m, 2H, H-5, H-6), 3.89 (m, 2H, H-4, H-6), 2.76–2.59 (m, 4H, CH_2(Lev)), 2.17 (s, 3H, CH_3(Lev)), 1.05–0.98 (2 s, 18H, Si(C(CH_3)$_3$)$_2$); ^{13}C NMR (100 MHz, $CDCl_3$): δ 206.6, 173.3 (CO), 157.4 (q, $^2J_{C,F}$ = 37.5 Hz, *C*OCF$_3$), 115.6 (q, $^1J_{C,F}$ = 285.9 Hz, CO*C*F$_3$), 91.2 (C-1), 75.0 (C-4), 72.5 (C-3), 66.6, 66.5 (C-5, C-6), 52.8 (C-2), 38.0 (CH_2(Lev)), 29.6 (CH_3(Lev)), 28.0 (CH_2(Lev)), 27.3, 26.8 (Si(C(CH_3)$_3$)$_2$), 22.7, 19.9 (Si(*C*(CH_3)$_3$)$_2$); ^{19}F NMR (376 MHz, $CDCl_3$): δ −76.0 (s, 3F); HR MS: *m/z*: calcd for $C_{21}H_{34}F_3O_8NSiNa$: 536.1898; found: 536.1889 [M + Na]$^+$.

***O*-(2-deoxy-3-*O*-levulinoyl-4,6-*O*-di-*tert*-butylsilylene-2-trifluoroacetamido-α,β-D-glucopyranosyl) trichloroacetimidate (17):** Trichloroacetonitrile (2.5 mL, 25.2 mmol) and catalytic DBU (13 μL, 0.084 mmol) were added to a solution of **16** (862 mg, 1.68 mmol) in dry CH_2Cl_2 (9 mL). After stirring for 2.5 h at room temperature, the reaction mixture was concentrated to dryness. The residue was purified by silica gel chromatography (hexane-EtOAc 5:1 + 1 %Et_3N) to afford **17** as a white amorphous solid (1.030 g, 94 %). TLC (hexane-EtOAc 4:1) Rf 0.47. Data for α anomer: ^{1}H NMR (400 MHz, $CDCl_3$): δ 8.81 (s, 1H, NH), 6.76 (d, 1H, $J_{2,NH}$ = 8.2 Hz, NH(TFA)), 6.36 (d, 1H, $J_{1,2}$ = 3.6 Hz, H-1), 5.29 (dd, 1H, H-3), 4.41 (m, 1H, H-2), 4.19 (dd, 1H, H-6), 4.04 (m, 2H, H-4, H-5), 3.92 (t, 1H, H-6), 2.77–2.63 (m, 4H, CH_2(Lev)), 2.17 (s, 3H, CH_3(Lev)), 1.06–0.96 (2 s, 18H, Si(C(CH_3)$_3$)$_2$); ^{13}C NMR (100 MHz, $CDCl_3$): δ 205.6, 173.9 (CO), 160.1 (C = NH), 157.4 (q, $^2J_{C,F}$ = 37.8 Hz, *C*OCF$_3$), 115.4 (q, $^1J_{C,F}$ = 286.1 Hz, CO*C*F$_3$), 93.9 (C-1), 90.6 (CCl_3), 74.2 (C-4 or C-5), 72.2 (C-3), 69.1 (C-4 or C-5), 66.2 (C-6), 52.5 (C-2), 38.0 (CH_2(Lev)), 29.5 (CH_3(Lev)), 27.9 (CH_2(Lev)), 27.3, 26.7 (Si(C(CH_3)$_3$)$_2$), 22.6, 20.0 (Si(*C*(CH_3)$_3$)$_2$); ^{19}F NMR (376 MHz, $CDCl_3$): δ −76.1 (s, 3F).

1*H*,1*H*,2*H*,2*H*-perfluorodecyl 2-deoxy-3-*O*-levulinoyl-4,6-*O*-di-*tert*-butylsilylene-2-trifluoroacetamido-β-D-glucopyranoside (18): Following the general procedure, **17** (83 mg, 0.126 mmol) was glycosylated with 1*H*,1*H*,2*H*,2*H*-perfluoro-1-decanol (117 mg, 0.252 mmol). The crude mixture was purified by F-SPE. The fluorophilic fraction was further purified by silica gel chromatography (toluene-EtOAc 16:1) to give **18** as a colourless oil (82 mg, 68 %). TLC (toluene-EtOAc 12:1) Rf 0.34; $[\alpha]_D^{20}$ −23° (*c* 1.0, $CHCl_3$); ^{1}H NMR (400 MHz, $CDCl_3$): δ 6.98 (d, 1H, $J_{2,NH}$ = 8.8 Hz, NH(TFA)), 5.17 (t, 1H, H-3), 4.78 (d, 1H, $J_{1,2}$ = 8.3 Hz, H-1), 4.23 (dd, 1H, $J_{5,6a}$ = 5.1 Hz, $J_{6a,6b}$ = 10.3 Hz, H-6a), 4.12 (m, 1H, *C*H_2CH$_2$C$_8$F$_{17}$), 3.96–3.91 (m, 3H, H-2, H-4, H-6b), 3.81 (m, 1H, *C*H_2CH$_2$C$_8$F$_{17}$), 3.53 (m, 1H, H-5), 2.78–2.58 (m, 4H, CH_2(Lev)), 2.38 (m, 2H, CH$_2$*C*H_2C$_8$F$_{17}$), 2.16 (s, 3H, CH_3(Lev)), 1.06–0.98 (2 s, 18H, Si(C(CH_3)$_3$)$_2$); ^{13}C NMR (100 MHz, $CDCl_3$): δ 206.4, 173.1 (CO), 157.7 (q, $^2J_{C,F}$ = 37.4 Hz, *C*OCF$_3$), 115.6 (q, $^1J_{C,F}$ = 286.1 Hz, CO*C*F$_3$), 100.7 (C-1), 74.7 (C-4), 73.5 (C-3), 70.9 (C-5), 66.1 (C-6), 62.0 (*C*H$_2$CH$_2$C$_8$F$_{17}$), 54.9 (C-2), 38.1 (CH_2(Lev)), 31.3 (CH$_2$*C*H$_2$C$_8$F$_{17}$), 29.4 (CH_3(Lev)), 28.0 (CH_2(Lev)), 27.3, 26.8 (Si(C(CH_3)$_3$)$_2$), 22.6, 19.9 (Si(*C*(CH_3)$_3$)$_2$); ^{19}F NMR (376 MHz, $CDCl_3$): δ −76.4 (s, 3F), −80.9 (t, 3F), −113.5 (m, 2F), −122.0 (m, 6F), −122.8 (m, 2F), −123.7 (m, 2F), −126.2 (m, 2F); HR MS: *m/z*: calcd for $C_{31}H_{37}F_{20}O_8NSiNa$: 982.1861; found: 982.1849 [M + Na]$^+$.

1*H*,1*H*-perfluorononyl 2-deoxy-3-*O*-levulinoyl-4,6-*O*-di-*tert*-butylsilylene-2-trifluoroacetamido-α,β-D-glucopyranoside (19): Following the general procedure, **17** (88 mg, 0.134 mmol) was glycosylated with 1*H*,1*H*-perfluoro-1-nonanol (121 mg, 0.268 mmol). The crude mixture was purified by F-SPE and the fluorophilic fraction was further purified by silica gel chromatography (hexane-EtOAc 7:1) to give compound **19** as a 1:1.7 α/β mixture (colourless oil, 58 mg, 46 %). TLC (toluene-EtOAc 12:1) Rf 0.43 and 0.36; Data for the β anomer: ^{1}H NMR (400 MHz, $CDCl_3$): δ 6.94 (d, 1H, $J_{2,NH}$ = 8.7 Hz, NH(TFA)), 5.23 (t, 1H, H-3), 4.91 (d, 1H, $J_{1,2}$ = 8.3 Hz, H-1), 4.28–3.87 (m, 6H, C*H*$_2$C$_8$F$_{17}$, H-2, H-4, H-6), 3.55 (m, 1H, H-5), 2.76–2.58 (m, 4H, CH_2(Lev)), 2.17 (s, 3H, CH_3(Lev)), 1.06–0.98 (2 s, 18H, Si(C(CH_3)$_3$)$_2$); ^{13}C NMR (100 MHz, $CDCl_3$): δ 206.4, 172.9 (CO), 157.5 (*C*OCF$_3$), 100.6 (C-1), 74.6 (C-4), 73.1 (C-3), 71.0 (C-5), 65.9 (C-6), 55.0 (C-2), 38.1 (CH_2(Lev)), 29.5 (CH_3(Lev)), 28.0 (CH_2(Lev)), 27.3, 26.8 (Si(C(CH_3)$_3$)$_2$), 22.6, 19.9 (Si(*C*(CH_3)$_3$)$_2$); Diagnostic peaks α anomer: ^{1}H NMR (400 MHz, $CDCl_3$): δ 6.73 (d, 1H, $J_{2,NH}$ = 8.7 Hz, NH(TFA)), 5.17 (t, 1H, H-3), 5.03 (d, 1H, $J_{1,2}$ = 3.7 Hz, H-1); ^{13}C NMR (100 MHz, $CDCl_3$): δ 205.8, 173.5 (CO), 97.7 (C-1), 71.9 (C-3), 67.6 (C-5), 52.5 (C-2); HR MS: *m/z*: calcd for $C_{30}H_{35}F_{20}O_8NSiNa$: 968.1705; found: 968.1689 [M + Na]$^+$.

1*H*,1*H*,2*H*,2*H*-perfluorodecyl *O*-(2-deoxy-3-*O*-levulinoyl-4,6-*O*-di-*tert*-butylsilylene-2-trifluoroacetamido β-D-glucopyranosyl)-(1 → 4)- *O*-(2-*O*-benzoyl-3-*O*-benzyl-6-*O*-pivaloyl-β-D-glucopyranosyl)-(1 → 3)-2-deoxy-4,6-*O*-di-*tert*-butylsilylene-2-trifluoroacetamido-β-D-glucopyranoside (20): As described in the general procedure, compound **18** (79 mg, 0.082 mmol) was delevulinoylated to afford the corresponding Lev-deprotected compound (64 mg). ESI MS: *m/z*: calcd for $C_{26}H_{30}F_{20}NO_6Si$: 860.2; found: 860.4 [M−H]$^-$.

The acceptor monosaccharide (63 mg, 0.073 mmol) was glycosylated with donor **3** (128 mg, 0.183 mmol) to afford the corresponding disaccharide (94 mg). ESI MS: *m/z*: calcd for $C_{56}H_{69}F_{20}N_2O_{15}Si$: 1417.4; found: 1417.6 [M + NH_4]$^+$.

This disaccharide (93 mg, 0.066 mmol) was submitted to the general procedure for Lev deprotection to yield the corresponding 4-OH disaccharide (80 mg). ESI MS: *m/z*: calcd for $C_{51}H_{58}F_{20}NO_{13}Si$: 1300.3; found: 1301.0 [M−H]$^-$.

Finally, the disaccharide acceptor (79 mg, 0.061 mmol) was glycosylated with donor **17** (100 mg, 0.152 mmol), as described in the general procedure. After this first glycosylation reaction, TLC and NMR analysis showed the presence of an important amount of unreacted glycosyl acceptor. For this reason, a second glycosylation cycle with additional donor **17** (100 mg, 0.152 mmol) was performed. The crude product was purified by silica gel column chromatography (toluene-EtOAc 16:1) to give pure trisaccharide **20** as a white amorphous solid (68 mg, 46 % from **18**, 4 steps). TLC (toluene-EtOAc 8:1) Rf 0.46; $[\alpha]_D^{20}$ −13° (*c* 1.0, $CHCl_3$); ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.94–7.40 (m, 5H, Ar), 7.36 (d, 1H, $J_{2,NH}$ = 9.2 Hz, NHTFA C), 7.17 (m, 5H, Ar), 6.66 (d, 1H, $J_{2,NH}$ = 7.6 Hz, NHTFA A), 5.20 (dd, 1H, $J_{1,2}$ = 7.0 Hz, $J_{2,3}$ = 8.2 Hz, H-2B), 5.01 (m, 2H, H-3C, H-1B), 4.76 (m, 2H, H-1A, CH_2(Bn)), 4.62 (d, 1H, J = 11.3 Hz, CH_2(Bn)), 4.58 (d, 1H, $J_{1,2}$ = 8.3 Hz, H-1C), 4.33 (br d, 2H, H-6B), 4.17–3.95 (m, 5H, C*H*$_2$CH$_2$C$_8$F$_{17}$, H-3A, H-2C, 2 × H-6A or C), 3.92–3.78 (m, 4H, H-6A or C, H-4A, H-4B, H-4C), 3.75–3.71 (m, 2H, C*H*$_2$CH$_2$C$_8$F$_{17}$, H-3B), 3.55–3.34 (m, 5H, H-6A or C, H-5B, H-2A, H-5A, H-5C), 2.77–2.59 (m, 4H, CH_2(Lev)), 2.33 (m, 2H, CH$_2$*C*H$_2$C$_8$F$_{17}$), 2.18 (s, 3H, CH_3(Lev)), 1.26 (s, 9H, C(CH_3)$_3$), 1.06–0.91 (4 s, 36H, Si(C(CH_3)$_3$)$_2$); ^{13}C NMR (100 MHz, $CDCl_3$): δ 206.1, 179.1, 172.8, 165.2 (CO), 157.6 (*C*OCF$_3$), 138.0–127.6 (Ar), 115.7 (CO*C*F$_3$), 100.8 (C-1C), 99.8 (C-1A), 98.5 (C-1B), 80.2 (C-3B), 77.1 (C-3A, C-4B or C), 74.8, 74.73, 74.67 (CH_2(Bn), C-4A, C-4B or C), 73.9, 73.7, 73.5 (C-2B, C-3C, C-5B), 71.0, 70.7 (C-5A, C-5C), 66.1, 65.7 (C-6A, C-6C), 62.3, 61.8 (C-6B, *C*H$_2$CH$_2$C$_8$F$_{17}$), 57.5 (C-2A), 54.9 (C-2C), 39.0 (*C*(CH_3)$_3$), 38.1 (CH_2(Lev)), 31.3 (t, CH$_2$*C*H$_2$C$_8$F$_{17}$), 29.5 (CH_3(Lev)), 27.9 (CH_2(Lev)), 27.4, 27.3, 27.1, 26.8, 26.7 (C(*C*H$_3$)$_3$), 22.7, 22.6, 19.9, 19.8 (Si(*C*(CH_3)$_3$)$_2$); ^{19}F NMR (376 MHz, $CDCl_3$): δ −75.8, −75.9 (2 s, 6F), −80.8 (t, 3F), −113.5 (m, 2F), −121.9 (m, 6F), −122.7 (m, 2F), −123.6 (m, 2F), −126.1 (m, 2F); HR MS: *m/z*: calcd for $C_{72}H_{91}F_{23}O_{20}N_2Si_2Na$: 1819.5229; found: 1819.5199 [M + Na]$^+$.

1*H*,1*H*,2*H*,2*H*-perfluorodecyl 2-deoxy-3-*O*-levulinoyl-2-phthalimido-4,6-*O*-di-*tert*-butylsilylene-β-D-glucopyranoside (21): Following the general procedure, **4** (132 mg, 0.191 mmol) was glycosylated with 1*H*,1*H*,2*H*,2*H*-perfluoro-1-decanol (177 mg, 0.381 mmol). The crude mixture was purified by F-SPE. The fluorophilic fraction was further purified by silica gel chromatography (toluene-EtOAc 16:1) to give **21** as a colourless oil (147 mg, 78 %). TLC (toluene-EtOAc 8:1) Rf 0.47; $[\alpha]_D^{20}$ −12° (*c* 1.0, $CHCl_3$); ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.86–7.72 (m, 4H, Ar), 5.71 (t, 1H, $J_{2,3}$ = $J_{3,4}$ = 9.9 Hz, H-3), 5.45 (d, 1H, $J_{1,2}$ = 8.4 Hz, H-1), 4.30–4.22 (m, 2H, H-2, H-6), 4.10–3.95 (m, 3H, H-4, H-6, *C*H$_2$CH$_2$C$_8$F$_{17}$), 3.79–3.70 (m, 2H, H-5, *C*H$_2$CH$_2$C$_8$F$_{17}$), 2.61–2.25 (m, 6H, CH_2(Lev), CH$_2$*C*H$_2$C$_8$F$_{17}$), 1.98 (s, 3H, CH_3(Lev)), 1.07–1.02 (2 s, 18H, Si(C(CH_3)$_3$)$_2$); ^{13}C NMR (100 MHz, $CDCl_3$): δ 205.6, 171.9 (CO), 134.0–123.3 (Ar), 98.5 (C-1), 75.7 (C-4), 72.3 (C-3), 70.7 (C-5), 66.2 (C-6), 61.9 (*C*H$_2$CH$_2$C$_8$F$_{17}$), 54.4 (C-2), 37.9 (CH_2(Lev)), 31.2 (t,

$CH_2CH_2C_8F_{17}$), 29.5 (CH_3(Lev)), 27.9 (CH_2(Lev)), 27.4, 26.9 (Si(C(CH_3)$_3$)$_2$), 22.7, 20.0 (Si(*C*(CH_3)$_3$)$_2$); ^{19}F NMR (376 MHz, $CDCl_3$): δ −80.8 (t, 3F), −113.4 (m, 2F), −122.0 (m, 6F), −122.7 (m, 2F), −123.6 (m, 2F), −126.1 (m, 2F); HR MS: *m/z*: calcd for $C_{37}H_{40}F_{17}O_9NNaSi$: 1016.2093; found: 1016.2083 $[M + Na]^+$.

1*H*,1*H*-perfluorononyl 2-deoxy-3-*O*-levulinoyl-2-phthalimido-4,6-*O*-di-*tert*-butylsilylene-α,β-D-glucopyranoside (22): Following the general procedure, **4** (89 mg, 0.129 mmol) was glycosylated with 1*H*,1*H*-perfluoro-1-nonanol (116 mg, 0.258 mmol). The crude mixture was purified by F-SPE to give compound **22** as a 1:4.5 α/β mixture (colourless oil, 40 mg, 32 %). TLC (toluene-acetone 11:1) Rf 0.53 and 0.44; Data for the β anomer: ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.86–7.72 (m, 4H, Ar), 5.75 (t, 1H, $J_{2,3} = J_{3,4} = 9.9$ Hz, H-3), 5.50 (d, 1H, $J_{1,2} = 8.4$ Hz, H-1), 4.31–3.99 (m, 6H, H-2, H-4, H-6, *CH*$_2$C$_8$F$_{17}$), 3.72 (m, 1H, H-5), 2.57–2.48 (m, 4H, CH_2(Lev)), 1.98 (s, 3H, CH_3(Lev)), 1.07, 1.01 (2 s, 18H, C(CH_3)$_3$); ^{13}C NMR (100 MHz, $CDCl_3$): δ 205.7, 171.8 (CO), 134.0–123.4 (Ar), 98.8 (C-1), 75.5 (C-4), 72.0 (C-3), 70.7 (C-5), 66.1 (C-6), 60.6 (t, $J_{C,F} = 25.8$ Hz, *CH*$_2$C$_8$F$_{17}$), 54.1 (C-2), 37.9 (CH_2(Lev)), 29.4 (CH_3(Lev)), 27.9 (CH_2(Lev)), 27.3, 26.8 (C(*C*H$_3$)$_3$), 22.7, 20.0 (*C*(CH_3)$_3$); Diagnostic peaks α anomer: ^{1}H NMR (400 MHz, $CDCl_3$): δ 6.52 (t, 0.23H, $J_{3,4} = 9.1$ Hz, H-3), 5.01 (d, 0.23H, $J_{1,2} = 3.6$ Hz, H-1), 4.54 (dd, 0.23H, $J_{2,3} = 11.6$ Hz, H-2); ^{13}C NMR (100 MHz, $CDCl_3$): δ 98.7 (C-1), 68.3 (C-3), 53.3 (C-2); HR MS: *m/z*: calcd for $C_{36}H_{38}NF_{17}O_9NaSi$: 1002.1937; found: 1002.1929 $[M + Na]^+$.

1*H*,1*H*,2*H*,2*H*-perfluorodecyl *O*-(2-*O*-benzoyl-3-*O*-benzyl-4-*O*-levulinoyl-6-*O*-pivaloyl-β-D-glucopyranosyl)-(1 → 3)-*O*-(2-deoxy-2-phthalimido-4,6-*O*-di-*tert*-butylsilylene-β-D-glucopyranosyl)-(1 → 4)-*O*-(2-*O*-benzoyl-3-*O*-benzyl-6-*O*-pivaloyl-β-D-glucopyranosyl)-(1 → 3)-*O*-(2-deoxy-2-phthalimido-4,6-*O*-di-*tert*-butylsilylene-β-D-glucopyranosyl)-(1 → 4)- *O*-(2-*O*-benzoyl-3-*O*-benzyl-6-*O*-pivaloyl-β-D-glucopyranosyl)-(1 → 3)-2-deoxy-2-phthalimido-4,6-*O*-di-*tert*-butylsilylene-β-D-glucopyranoside (23): Following the general procedures for deprotection of Lev groups and glycosylation, compound **21** (70 mg, 0.071 mmol) was successively transformed into the corresponding Lev-deprotected monosaccharide, glycosylated with donor **3** (117 mg, 0.167 mmol), delevulinoylated, glycosylated with donor **4** (102 mg, 0.148 mmol), Lev deprotected, treated with donor **3** (102 mg, 0.146 mmol), delevulinoylated, glycosylated with trichloroacetimidate **4** (82 mg, 0.118 mmol) and delevulinoylated to give the desired 3-OH pentasaccharide acceptor. All these steps were checked by ESI mass spectrometry. Final coupling with donor **3** (64 mg, 0.091 mmol), again following the general procedure for glycosylation, gave the crude hexasaccharide **23** after F-SPE. Purification by silica gel column chromatography (toluene-EtOAc 8:1) afforded pure hexasaccharide **23** as a white amorphous solid (53 mg, 24 % from **21**, 10 steps). TLC (toluene-EtOAc 7:1) Rf 0.26; $[\alpha]_D^{20} + 2^\circ$ (*c* 1, $CHCl_3$); ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.70–6.94 (m, 42H, Ar), 5.13 (t, 1H, $J_{3,4} = J_{4,5} = 9.3$ Hz, H-4F), 5.06–4.99 (m, 3H, H-2F, 2 × H-1A, C or E), 4.94 (d, 1H, $J_{1,2} = 8.4$ Hz, H-1A, C or E), 4.87 (m, 3H, H-2B, H-2D, H-1F), 4.65–4.52 (m, 5H, 2 × CH_2(Bn), H-1B, H-1D, H-3A, C or E), 4.41–4.32 (m, 4H, 2 × CH_2(Bn), 2 × H-3A, C or E), 4.25 (m, 2H, CH_2(Bn)), 4.18–3.99 (m, 8H, H-2A, H-2C, H-2E, 2 × H-6F, 2x H-6B or D, H-6 A, C or E), 3.95–3.63 (m, 9H, H-4A, H-4B, H-4C, H-4D, H-4E, 1 × *CH*$_2$CH$_2$C$_8$F$_{17}$, 3 × H-6 A, C or E), 3.63–3.55 (m, 3H, H–3F, H-5F, *CH*$_2$CH$_2$C$_8$F$_{17}$), 3.47–3.22 (m, 9H, H-5A, H-5C, H-5E, H-3B, H-3D, 2 × H-6 A, C or E, 2 × H-6B or D), 3.04 (m, 2H, H-5D, H-5B), 2.65–2.38 (m, 4H, CH_2(Lev)), 2.13 (m, 5H, $CH_2CH_2C_8F_{17}$, CH_3(Lev)), 1.24–1.15 (3 s, 3 × 9H, 3 × C(CH_3)$_3$), 1.04–0.84 (6 s, 54H, Si(C(CH_3)$_3$)$_2$); ^{13}C NMR (100 MHz, $CDCl_3$): δ 206.1, 178.3, 177.1, 171.1, 168.3, 167.0, 164.9, 164.80, 164.76 (CO), 138.1–123.2 (Ar), 100.0, 99.8, 99.7 (C-1B, C-1D, C-1F), 98.4, 97.4 (C-1A, C-1C, C-1E), 80.3, 80.2 (C-3B, C-3D, C-3F), 77.8–76.8, 75.3, 75.1 (C-3A, C-3C, C-3E, C-4A, C-4C, C-4E, C-4B, C-4D), 74.4–73.3 (3 × CH_2(Bn), C-2B, C-2D, C-2F, C-5B, C-5D), 72.7 (C-5F), 70.6, 70.3, 70.2 (C–5A, C–5C, C-5E), 69.3 (C-4F), 66.2–65.7 (C-6A, C–6C, C-6E), 62.3, 62.1 (C-6B, C-6D, C-6F), 61.5 (*CH*$_2$CH$_2$C$_8$F$_{17}$), 55.9, 55.8, 55.1 (C-2A, C-2C, C-2E), 38.9, 38.7 (*C*(CH_3)$_3$), 37.8 (CH_2(Lev)), 31.1 (t, $CH_2CH_2C_8F_{17}$), 29.7 (CH_3(Lev)), 27.8 (CH_2(Lev)), 27.4–26.9 (C(*C*H$_3$)$_3$, Si(C(*C*H$_3$)$_3$)$_2$), 22.51, 22.46, 22.3, 19.9, 19.83, 19.76 (Si(*C*(CH$_3$)$_3$)$_2$); ^{19}F NMR (376 MHz, $CDCl_3$): δ −80.7 (t, 3F), −113.6 (m, 2F), −122.0 (m, 6F), −122.7 (m, 2F), −123.8 (m, 2F), −126.1 (m, 2F); HR MS: *m/z*: calcd for $C_{156}H_{182}F_{17}O_{42}N_3NaSi_3$: 3199.1126; found: 3199.1047 $[M + Na]^+$.

1*H*,1*H*,2*H*,2*H*-perfluorodecyl *O*-(2-*O*-benzoyl-3-*O*-benzyl-4-*O*-levulinoyl-6-*O*-pivaloyl-β-D-glucopyranosyl)-(1 → 3)-*O*-(2-deoxy-2-phthalimido-4,6-di-*O*-sulfo-β-D-glucopyranosyl)-(1 → 4)-*O*-(2-*O*-benzoyl-3-*O*-benzyl-6-*O*-pivaloyl-β-D-glucopyranosyl)-(1 → 3)-*O*-(2-deoxy-2-phthalimido-4,6-di-*O*-sulfo-β-D-glucopyranosyl)-(1 → 4)-*O*-(2-*O*-benzoyl-3-*O*-benzyl-6-*O*-pivaloyl-β-D-glucopyranosyl)-(1 → 3)-2-deoxy-2-phthalimido-4,6-di-*O*-sulfo-β-D-glucopyranoside (24): An excess of $(HF)_n$·Py (74 μL, 2.8 mmol) was added at 0 °C under an argon atmosphere to a solution of **23** (45 mg, 14.2 μmol) in dry THF (1.5 mL). After 24 h at 0 °C, the mixture was diluted with CH_2Cl_2 and washed with H_2O and saturated $NaHCO_3$ solution. The organic layers were dried ($MgSO_4$), filtered and concentrated in vacuo to give the corresponding hexaol (39 mg) that was directly sulfated. TLC (toluene-acetone 2:1) Rf 0.35; ESI MS: *m/z*: calcd for $C_{132}H_{134}F_{17}N_3O_{42}Na$: 2779.8; found: 2780.0 $[M + Na]^+$.

The hexaol (14.2 μmol) and sulfur trioxide–trimethylamine complex (118 mg, 0.85 mmol) were dissolved in dry DMF (2.4 mL) and heated at 100 °C for 30 min using microwave radiation. Saturated $NaHCO_3$ solution (0.24 mL) and CF_3CH_2OH (60 μL) were added and the reaction mixture was purified by F-SPE. In this case, the F-SPE column was preconditioned with MeOH/H_2O 40:60 (20 mL). Nonfluorous compounds were eluted with MeOH/H_2O 40:60 (15 mL) and the fluorophilic elution was performed using 100 % MeOH (20 mL). The fluorous fraction was concentrated under reduced pressure and submitted to two additional sulfation cycles, at 120 °C for 30 min using microwaves. Finally, the crude mixture was purified by F-SPE (as described just before) and by silica gel column chromatography (EtOAc-MeOH-H_2O 31:5:3 → EtOAc-MeOH-H_2O 24:5:3). After elution from a Dowex 50WX2-Na^+ column (MeOH), **24** (12 mg, 25 %, 2 steps) was obtained as sodium salt. TLC (EtOAc-MeOH-H_2O 20:5:3) Rf 0.33; ^{1}H NMR (400 MHz, CD_3OD): δ 7.68–6.88 (m, 42H, Ar), 5.20 (d, 1H, $J_{1,2} = 8.5$ Hz, H-1A, C or E), 5.12 (m, 2H, H-4F, 1 × H-1A, C or E), 5.07–4.97 (m, 4H, H-1F, H-2F, 1 × H-1A, C or E, 1 × H-3A, C or E), 4.86 (m, 2H, H-2B, H-2D), 4.77–4.59 (m, 8H, 2 × CH_2(Bn), H-1B or D, 2 × H-3A, C or E, 3 × H-6A, C or E), 4.54 (d, 1H, $J_{1,2} = 7.6$ Hz, H-1B or D), 4.49–4.41 (m, 4H, 3 × CH_2(Bn), 1 × H-4A, C or E), 4.33–3.78 (m, 21H, 1 × CH_2(Bn), H-2A, H-2C, H-2E, 2 × H-6F, 2x H-6B or D, 3 × H-6 A, C or E, 2 × H-4 A, C or E, H-4B, H-4D, 1 × *CH*$_2$CH$_2$C$_8$F$_{17}$, H-3F, H-5F, H-5A, H-5C, H-5E), 3.63 (m, 3H, 1 × *CH*$_2$CH$_2$C$_8$F$_{17}$, 2x H-6B or D), 3.45 (m, 2H, H-3B, H-3D), 3.33 (m, 2H, H-5D, H-5B), 2.73–2.45 (m, 4H, CH_2(Lev)), 2.25–2.12 (m, 5H, $CH_2CH_2C_8F_{17}$, CH_3(Lev)), 1.24–1.18 (3 s, 3 × 9H, 3 × C(CH_3)$_3$); ^{13}C NMR (100 MHz, CD_3OD, selected data from HSQC experiment): δ 98.9, 98.8, 98.6 (C-1B, C-1D, C-1F), 96.8, 96.3 (C-1A, C-1C, C-1E), 80.5 (C-3F), 79.2 (C-3B, C-3D), 76.7, 76.1 (C-4A, C-4C, C-4E), 75.5, 75.3 (C-3A, C-3C, C-3E), 73.6 (CH_2(Bn), C-2B, C-2D, C-2F), 72.8 (C–5B, C–5D), 69.5 (C-4F), 67.0 (C-6A, C–6C, C-6E), 62.3, 61.6 (C-6B, C-6D, C-6F), 60.8 (*CH*$_2$CH$_2$C$_8$F$_{17}$), 55.8, 55.5 (C-2A, C-2C, C-2E), 36.9 (CH_2(Lev)), 30.0 ($CH_2CH_2C_8F_{17}$), 28.1 (CH_3(Lev)), 27.7 (CH_2(Lev)); ESI MS: *m/z*: calcd for $C_{132}H_{130}F_{17}N_3O_{60}S_6$: 808.1; found: 808.2 $[M-6Na + 2H]^{4-}$.

1*H*,1*H*,2*H*,2*H*-perfluorodecyl *O*-(3-*O*-benzyl-β-D-glucopyranosyl)-(1 → 3)-*O*-(2-acetamido-2-deoxy-4,6-di-*O*-sulfo-β-D-glucopyranosyl)-(1 → 4)-*O*-(3-*O*-benzyl-β-D-glucopyranosyl)-(1 → 3)-*O*-(2-acetamido-2-deoxy-4,6-di-*O*-sulfo-β-D-glucopyranosyl)-(1 → 4)-*O*-(3-*O*-benzyl-β-D-glucopyranosyl)-(1 → 3)-2-acetamido-2-deoxy-4,6-di-*O*-sulfo-β-D-glucopyranoside (2): Ethylene diamine (89 μL, 1.33 mmol) was added, under an argon atmosphere, to a solution of **24** (10 mg, 3.0 μmol) in a mixture of *n*-BuOH (1.2 mL) and 2,2,2-trifluoroethanol (0.2 mL). Then, the reaction mixture was subjected to microwave irradiation for 120 min at 120 °C (4 cycles, 30 min each). The reaction vessel was cooled, and the mixture was concentrated to dryness. The residue was disolved in MeOH/CF_3CH_2OH (2.0 mL/0.1 mL) and an

aqueous solution of NaOH (4 M, 223 μL) was added. After stirring for 72 h at room temperature, the reaction mixture was neutralized with Amberlite IR-120 (H^+) resin, filtered, and concentrated to give the desired amine intermediate. Triethylamine (16 μL, 0.12 mmol) and acetic anhydride (17 μL, 0.18 mmol) were added to a cooled (0 °C) solution of this amine derivative in MeOH (1.5 mL). After stirring for 3 h at room temperature, additional triethylamine (32 μL, 0.24 mmol) and acetic anhydride (34 μL, 0.36 mmol) were added at 0 °C to complete the reaction. After further stirring for 4.5 h at room temperature, Et_3N (0.2 mL) was added and the mixture was concentrated to dryness. The residue was then purified by silica gel column chromatography (EtOAc-MeOH-H_2O 24:5:3 → EtOAc-MeOH-H_2O 18:5:3) and finally eluted from a Dowex 50WX2-Na^+ column (MeOH) to obtain **2** as sodium salt (5.7 mg, 78 %, white amorphous solid). TLC (EtOAc-MeOH-H_2O 16:5:3) Rf 0.24; ^{1}H NMR (400 MHz, CD_3OD): δ 7.52–7.24 (m, 15H, Ar), 4.96–4.85 (m, 6H, 3 × CH_2(Bn)), 4.83–4.74 (m, 3H, 2 × H-1A, C or E, 1 × H-6A, C or E), 4.65–4.46 (m, 6H, 1 × H-1A, C or E, 2 × H-6A, C or E, H-1B, H-1D, H-1F), 4.29–4.09 (m, 7H, 3 × H-6A, C or E, H-4A, H-4C, H-4E, 1 × $CH_2CH_2C_8F_{17}$), 4.00–3.57 (m, 22H, H-2A, H-2C, H-2E, H-3A, H-3C, H-3E, H-4B, H-4D, 1 × $CH_2CH_2C_8F_{17}$, H-3B, H-3D, H-6B, H-6D, H-6F, H-5A, H-5C, H-5E, H-5B, H-5D), 3.52 (m, 3H, H-2B, H-2D, H-4F), 3.43 (m, 1H, H-2F), 3.33 (m, 1H, H-3F), 3.26 (m, 1H, H-5F), 2.51 (m, 2H, $CH_2CH_2C_8F_{17}$), 2.02–1.98 (3 s, 9H, NHAc); ^{13}C NMR (100 MHz, CD_3OD, selected data from HSQC experiment): δ 103.0 (C-1B, C-1D, C-1F), 100.4–98.6 (C-1A, C-1C, C-1E), 84.2 (C-3F), 80.3 (C-3B, C-3D), 78.4 (C-3A, C-3C, C-3E), 76.4 (C-5F), 74.4 (C-4A, C-4C, C-4E), 73.7–71.8 (CH_2(Bn)), 73.3–73.1 (C-2B, C-2D, C-2F), 69.1 (C-4F), 67.3 (C-6A, C–6C, C-6E), 61.2 ($CH_2CH_2C_8F_{17}$), 60.6–59.7 (C-6B, C-6D, C-6F), 55.2 (C-2A, C-2C, C-2E), 30.6 ($CH_2CH_2C_8F_{17}$), 21.8 (NHAc); ^{19}F NMR (376 MHz, CD_3OD): δ −82.3 (t, 3F), −114.3 (m, 2F), −122.8 (m, 6F), −123.7 (m, 2F), −124.5 (m, 2F), −127.3 (m, 2F); HR MS: *m/z*: calcd for $C_{73}H_{86}F_{17}O_{49}N_3S_6Na_4$: 1197.5992; found: 1197.6047 $[M-2Na]^{2-}$.

4.2. Fluorescence polarization competition assays

Fluorescence polarization measurements were performed in 384-well microplates (black polystyrene, non-treated, Corning), as previously reported [23]. The fluorescence polarization was recorded using a TRIAD multimode microplate reader (from Dynex), with excitation and emission wavelengths of 485 and 535 nm, respectively. The fluorescent probe (a fluorescein labelled heparin-like hexasaccharide previously prepared in our lab) [65] was dissolved in PBS buffer (10 mM, pH 7.4). Recombinant human MK and FGF-2 (Peprotech) were dissolved in PBS buffer (10 mM, pH 7.4) containing 1 % BSA (bovine serum albumin). 1 mM stock solutions of compounds **1**, **2**, **10** and **11** were prepared in PBS/DMSO 9:1 (v/v) and serial dilutions were then performed in PBS buffer (10 mM, pH 7.4).

For the determination of the IC_{50} values, we recorded the fluorescence polarization from wells containing 20 μL of protein solution (125 nM MK solution or 194 nM FGF-2 solution) and 10 μL of a 40 nM probe solution in the presence of 10 μL of inhibitor solution, with concentrations ranging from 1 mM to 0.1 μM. The microplate was shaked in the dark for 5 min, before reading. The total sample volume in each well was 40 μL and the final buffer composition was PBS + 0.5 % BSA. The final concentrations of fluorescent probe and MK/FGF-2 in each well were 10 nM and 63/97 nM, respectively, while the final inhibitor concentration ranged from 250 μM to 25 nM. The average polarization values of three replicates were plotted against the logarithm of inhibitor concentration. Two control samples were included in the competition experiment. The first one only contained fluorescent probe and afforded the expected minimum polarization value for 100 % inhibition; the second one contained MK/FGF-2 and probe, in the absence of inhibitor, and gave the maximum polarization value corresponding to 0 % inhibition. Blank wells contained 20 μL of protein solution and 20 μL of a 50 μM inhibitor solution and their measurements were subtracted from all values. The curve was fitted to the equation for a one-site competition: $y = A_2 + (A_1-A_2)/[1 + 10\hat{}(x\text{-}logIC_{50})]$ where A_1 and A_2 are the maximal and minimal values of polarization, respectively, and IC_{50} is the inhibitor concentration that results in 50 % inhibition. At least three independent experiments were carried out for each IC_{50} calculation.

4.3. NMR experiments

NMR experiments were recorded in a 600 MHz Bruker Avance III instrument, equipped with a cryoprobe (600, QCI Cryo 5 m (1H/19F 15 N/13C) for ^{1}H, ^{15}N, ^{13}C, and ^{19}F; with 2H decoupling). We have assigned the ^{1}H and ^{13}C spectra using the standard methodology based on ^{1}H and ^{13}C NMR methods at natural abundance.

NMR experiments were recorded using the manufacturer pulse sequences: 1D, 1D-TOCSY [72] or 1D-NOESY [73], 2D COSY with double quantum filter with pulsed field gradient for coherence selection [74] TOCSY, with using dipsi2 sequence in the mixing period; NOESY was implemented in phase-sensitive mode with two 180° pulses flanked with gradients in the mixing time [73]. Both, TOCSY and NOESY were acquired with a z-filter to suppress interferences of zero quantum coherence [75]. NOESY experiments were preferentially acquired using non-uniform sampling at 100 % to minimize periodic interferences from the environment due to thermal instability. HSQC were generally acquired using echo/antiecho-TPPI acquisition with an increase of sensitivity [76], Chirp 180° pulses in ^{13}C, and gradients during back inept. All the off resonance ROESY were acquired with the EASY-ROESY sequence [77] with 4 titled angles and five mixing times. HSQC sequence includes an edition module [78].

STD-NMR experiments have been performed using home-written sequences with solvent suppression using watergate sequences both 3–9-19 or w5 sequences. To avoid the influence of the proton relaxation rates, we use STD_0 calculated from the evolution of the STD with the saturation time (tsat), obtained from the fitted equation STD-AF(t) = a (1-exp(-bt)), where the parameter a represents the asymptotic maximum of the STD build-up curve (STD_{max}), b is a rate constant related to the relaxation properties of a given proton that measures the speed of the STD build-up (k_{sat}), and t, is the saturation time (t_{sat}). Thus, the STD_0 values were obtained as the product of the ab coefficients [79].

4.4. Cell proliferation assays

NIH3T3 cells were cultured in Dulbecco's modified Eagle's medium (DMEM) (Sigma-Aldrich) supplemented with 10 % fetal bovine serum (Sigma-Aldrich) and 10 mL/L of an antibiotic solution mix with Penicillin (100 U/mL) and Streptomycin (10 mg/mL) (Sigma-Aldrich). MK protein (20 μg/mL, from Peprotech) and the different compounds (2 mM) were combined 1: 1 before adding them to cells to get final concentrations of 100 ng/mL and 10 μM, respectively. 10.000 cells were plated on 30 mm culture dishes at day 0, and cells were counted at days 1 to 4 in a Neubauer chamber on an inverted microscope (Leika).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bioorg.2023.106929.

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